

Global Financial Markets

(student Version, August 2, 2025)

EMBA 60 Global

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List Of Books and Readings

- Books: Contemporary Financial Intermediation by Greenbaum, Thakor and Boot, *Elsevier*, 2019.
- The Purpose of Banking, Thakor, Oxford University Press.
- Readings:
 - Hydro case
 - Bird in Hand Bank Case

Goals

- To provide a description of the global and U.S. financial services landscape. To help you understand financial intermediaries, markets and contracts, and give you the **appropriate “vocabulary”**.
- ***The Main Street-Wall Street Connection:*** to understand the economic interaction between the real and financial sectors and the services provided by the financial sector, including the financing of start-ups.
- To examine the 2007-09 **financial crisis** and comprehend the multi-faceted role of central banks.
- To understand the role of **securitization and shadow banking** in the modern economy.
- To understand **major risks**, how they interact and how they should be managed.

Schedule: First Day (September 5, 2025, 1:30 -6 pm. Afternoon)

- Module I: The Financial Market Landscape
- Module II: Economic Services Provided by Banks and the Financial System
 - Reflection Exercise
- Module III: Securitization and Shadow Banking
 - ➡ Group Exercise #1 : Case Studies in Financial Engineering

Schedule: Second Day

(September 6) 8:00 am – 6:00 pm

- Module III: Securitization (Continued)
 - ➡ Group Exercise #2: Lone Star Bank
- Module IV: The Financial Crisis
- Module V: Risk Management
 - ➡ Group Exercise #3: Hydro Case
- Module VI: Banking and Higher Purpose
 - ➡ Group Exercise #4: Bird in Hand Bank Case

Schedule: Second Day (cont.)

(September 6, 2025) 8 am – 6:00 pm

- Module VII: Raising Financing

- ➔ GRADING FOR MY PORTION:
- ➔ IN-CLASS GROUP PROJECTS: 40%
- ➔ FINAL GROUP BOOK PROJECT: 20%
- ➔ FINAL EXAM (INDIVIDUAL) : 40%

BOOK PROJECT: At the end of the course, write a 3-5 page report on *The Purpose of Banking* book, summarizing its most important takeaways for you. This is a group assignment. This report will be due on the same day as your individual final exam, which will be due 2 weeks after class is over

“Don’t it always seem to go that you don’t know what you’ve got ‘til it’s gone?”

Joni Mitchell

Module I:

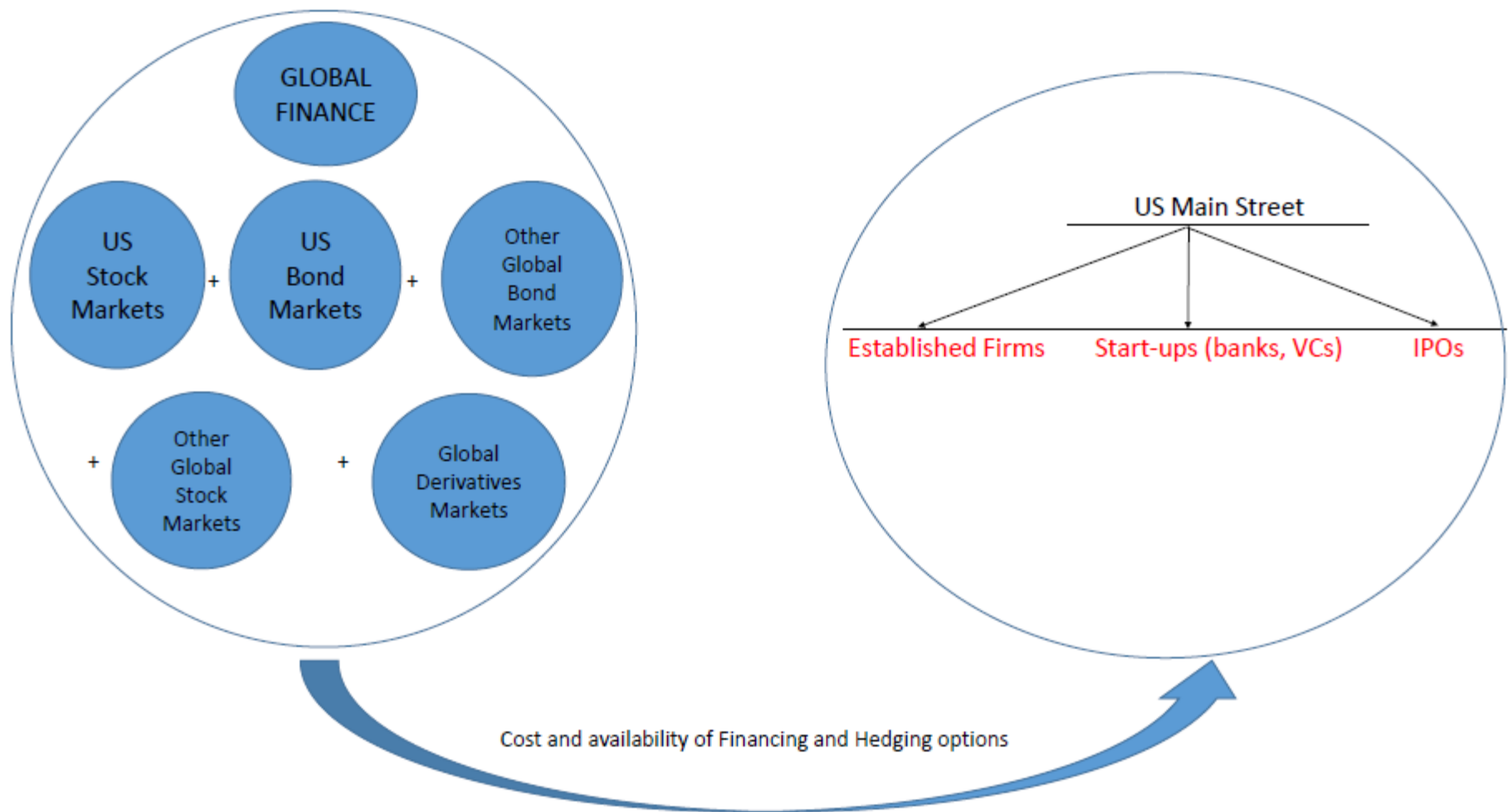
Financial Market Landscape

Contemporary Financial Intermediation, Chapter 2.
The Nature and Variety of Financial Intermediation

KEY QUESTIONS

- How global is Finance?
- What are financial intermediaries (F.I.s)?
- Why do we have F.I.s? What do F.I.s do that could not be done without them?
- What services do F.I.s provide?
- What is the scope of the financial services industry?
- How do we classify different types of F.I.s?

There is nothing more global than Finance



2024 GLOBAL ECONOMIC AND MARKET DATA

- Global stock market capitalization: \$126.7 trillion
- US stock market capitalization accounted for 50.2% of global capitalization.
- 63 of the top 100 market cap companies are HQed in the US and account for 76.7% of market cap of top 100!
- “Magnificent Seven” (Apple, Microsoft, Nvidia, Alphabet, Amazon, Meta and Tesla) had July 2025 market cap of \$19 trillion which > total market capitalization of :
(China+ Canada+ UK) and also that of (Canada+ Japan + UK+ Germany)

MORE 2024 GLOBAL DATA

- Global bond market: \$145 trillion
- Global GDP: \$110 trillion
- US: \$29.2 trillion
- EU: \$19.4 trillion
- China: \$18.7 trillion
- Thus, the US accounts for almost 27% of global GDP, more than 50% of global stock market capitalization, and over 30% of global consumption.

WHAT ACCOUNTS FOR CONTINUED US DOMINANCE?

- Very deep and liquid capital markets
- **Excellent banking system**
- Thriving entrepreneurship, low risk aversion, excellent financial ecosystem (VCs, PE firms, private credit etc.) to support entrepreneurship
- Innovation, powerhouse universities, research
- Culture of free markets , competition and meritocracy

WHY DO BANKS MATTER?

- Because they help to lower distortions due to various contracting frictions that impede the flow of capital from savers to investors-- asymmetric information, moral hazard--and they create a variety of financial options.
- Lower frictions and more options mean more efficient deployment of capital.

THREE BIGGEST CONTRIBUTIONS TO ECONOMICS IN LAST 50 YEARS:

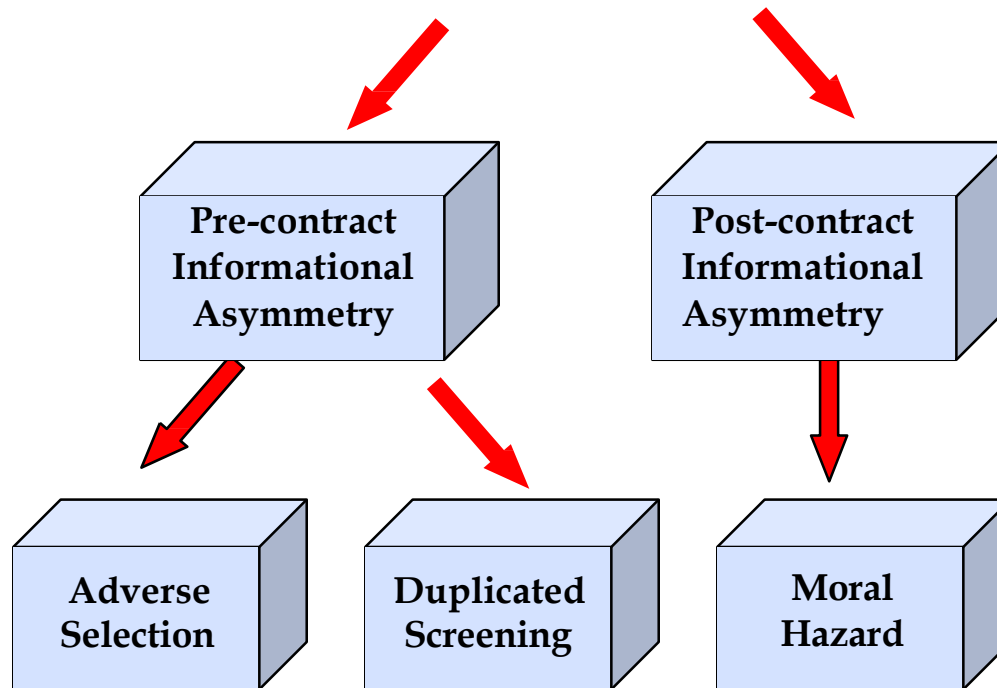
- Incentives (moral hazard) and optimal contracting
- Asymmetric information (adverse selection)
- Options
- This course touches on all three...

WHAT DO BANKS AND OTHER FI DO?

- When You Put Money in the Bank annndddd It's Gone - SOUTH PARK - YouTube

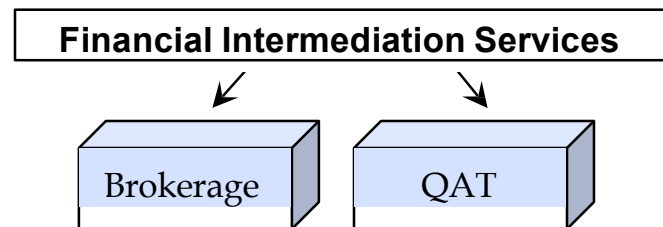
Why do we have F.I.s?

- Key informational problems



Why Do We Have F.I.s?

- Because they help us resolve informational problems.
- Information-based financial services produced by F.I.s:
 - ➔ “**Brokerage**” – joining unfamiliar but well-suited and complementary transactors in financial claims
 - ➔ “**Qualitative asset transformation**” (QAT) – allocating credit presumably to its highest and best uses while reconfiguring the attributes of the financial claims held by its clienteles



The Brokerage Function

- An important component of financial intermediation services:
 - ➔ Bringing together transactors in financial claims with complementary needs
 - ➔ A broker's stock-in-trade is information
- Two advantages of a broker as an information processor:
 - ➔ Specific skills in interpreting subtle (not readily observable) signals
 - ➔ Cross-sectional (across customers) and intertemporal (through time) information reusability

The Brokerage Function

- The broker helps resolve problems that arise from pre-contract informational asymmetry
 - ➔ *Pre-contract*: adverse selection and duplicated screening
 - ➔ *Post-contract*: moral hazard

ADVERSE SELECTION

- Akerlof (1970), used car example (Chapter 1)
- An example in lending:
 - ➔ Borrowers wish to overstate their credit worthiness → lenders increase loan interest rate to compensate higher credit risk → low-credit-risk borrowers drop out → lenders left with high-credit-risk borrowers
- *Solution:* F.I.s perform the brokerage function of credit analysis to sort out borrowers of different credit risks and minimize adverse selection problems
- Credit analysis is a form of “screening.” FIs specialize in this.

Duplicated Screening:

- Why can't non-FIs do screening? That is, as in P2P lending, why can't savers directly screen borrowers?
- They can, but FI can do better. How? By avoiding duplicated screening.
- An example of marriage broker
 - ➡ Before there is a marriage broker:
 - ❖ 100 men and 100 women searching for the “perfect” marriage partner, $x = 100$
 - ❖ Cost of evaluating one person, $c = \$25$
 - ❖ Total evaluation cost
 - $= \$25 \times 2 \times 100 \times 100$
 - $= \$500,000$
 - $= 2cx^2$

DUPLICATED SCREENING (CONT'D.)

➡ Now, enter the broker who needs to evaluate each man and woman only once:

❖ Total evaluation cost

$$\begin{aligned} &= \$25 \times 2 \times 100 \\ &= \$5,000 \\ &= 2cx \end{aligned}$$

➡ Savings, $S = \$495,000 = 2cx(x - 1)$

DUPLICATED SCREENING

- Continue with the example
 - ➔ The savings with broker arise due to *cross-sectional reusability of information*
 - ❖ A peculiarity of information: its use does not result in its consumption

DUPLICATED SCREENING (CONT'D.)

Role of Expertise:

➔ Assume the broker has a relative advantage in information evaluation:

❖ C_b = broker's evaluation cost

❖ C_o = others evaluation cost

❖ $C_o > C_b$

❖ Savings due to the broker,

$$S = 2x[C_o - C_b],$$

which grows with the gap, $C_o - C_b$

➔ Also, savings increase as the grid size x increases

INFORMATION REUSABILITY

- Two types of information reusability
 - ➔ *Cross-sectional*: the same information can be utilized across different users
 - ➔ *Intertemporal*: the same information can be reused through time
- To summarize:
 - ➔ For a given attribute, the larger the grid, the more compelling the need for the broker
 - ➔ For a given size grid, the less readily observable the attribute (the more subtle the signal), the more important the skills and reputation of the broker, e.g., phone books vs. security analysts or loan officers

IMPORTANT INSIGHT OF SIMPLE EXAMPLE

- Economies of scale in financial intermediation are unbounded...thus, these are “natural monopolies”.
- Thinks of CRAs, Amazon, Google, You Tube, Apple...
- What other business ideas can you come up with that exploit this feature of information reusability? (Answer this and become very rich!!)

MORAL HAZARD

- Moral hazard: situations in which the incentives of principal and agent diverge (Chapter 1)
 - ➔ One party of the transaction can take actions at the expense of the other party
 - ➔ These actions are “hidden” from the injured party and cannot be directly controlled or prevented
- Examples:
 - ➔ In insurance, the insured may underinvest in costly efforts to prevent adverse outcomes
 - ➔ In banking, borrowers may choose excessively risky projects

MORAL HAZARD

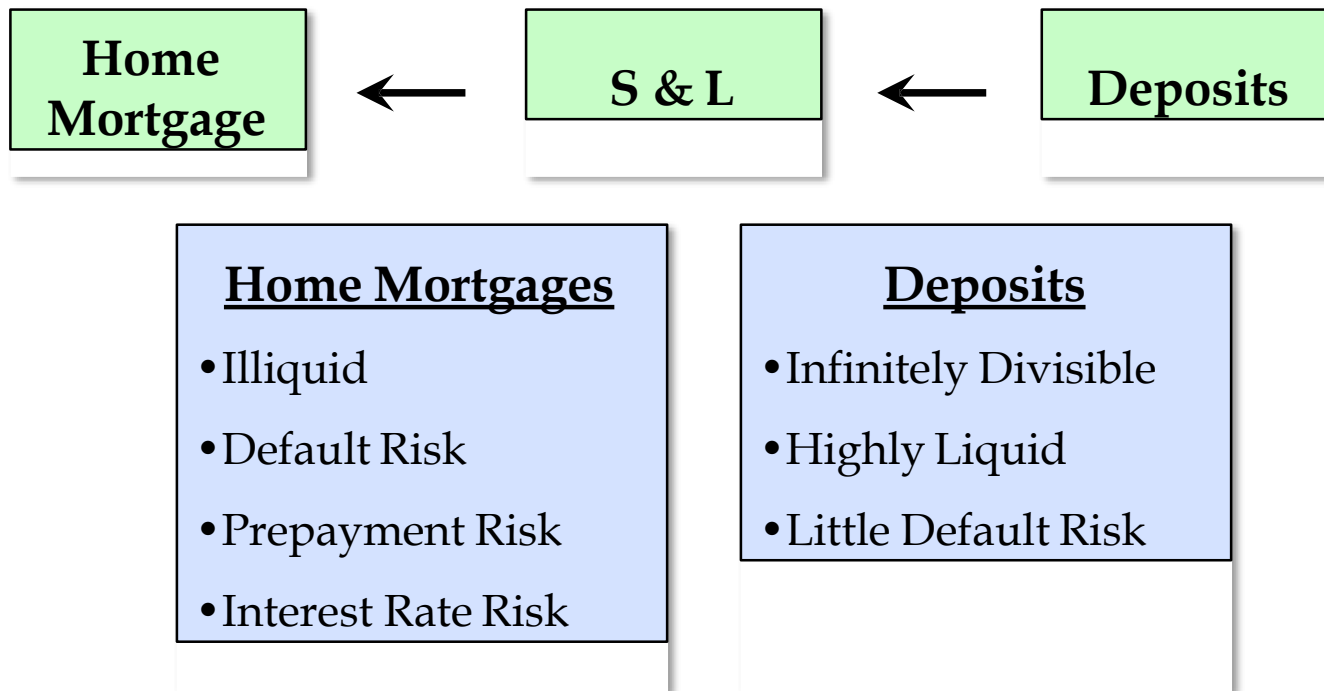
- Solution: F.I.s' special skills in monitoring attenuate moral hazard
- Examples:
 - ➔ Banks periodically examine borrowers' business and financial condition and intervene in operating strategy if necessary
 - ➔ Insurance contracts use ex post pricing adjustments to deter moral hazard
 - ➔ Venture capitalists threaten entrepreneurs with transfer of control

QUALITATIVE ASSET TRANSFORMATION (QAT)

- F.I.s need not expose their own capital to risk to perform the brokerage function
 - ➡ The processing of risk is not central to the production of brokerage services
- But, some financial intermediation activities requires F.I.s to expose their capital to risk – this is when F.I.s perform *Qualitative Asset Transformation* (QAT)

AN EXAMPLE OF QAT

- QAT transforms the nature of financial claims



QAT AND RISK

- Every QAT requires a *mismatch* on the F.I.'s balance sheet, and this mismatch is an inherent source of risk (**What mismatch did SVB have when it failed in 2023?**)
- Such a mismatch reflects an acceptance of some type of risk exposure by the F.I., for example:
 - ➡ Holding assets of *longer duration* than its liabilities
 - ➡ Enhancing *liquidity* of clients' claim by holding assets less liquid than liabilities

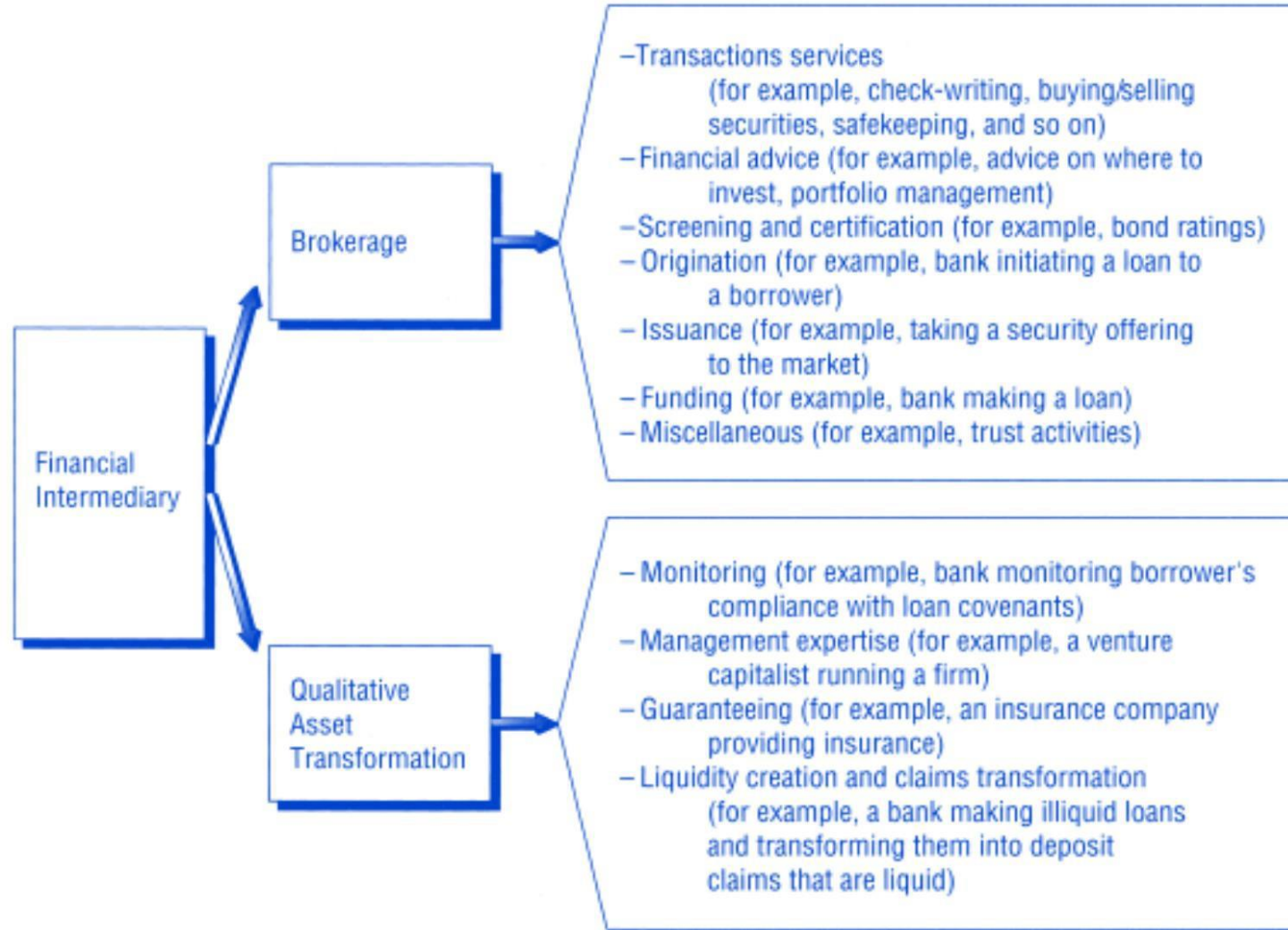
QAT AND RISK

- Hence, risk is integral to QAT
- Three alternatives available to F.I.s to manage this risk:
 - ➔ Accept risk passively
 - ➔ Diversify and reduce risk
 - ➔ Shift risk to others through the use of claims such as swaps, forward contracts, futures, options, swaptions, etc., that are traded in global financial markets.



The more risk-shifting, the more the QAT reverts to brokerage

SERVICE PROVIDED BY F.I.S



THE VARIETY OF F.I.S

- Depository F.I.s
 - ➔ F.I.s that finance (at least partly) with deposits: commercial banks, thrifts, credit unions
- Nondepository F.I.s
 - ➔ F.I.s that do not finance with deposits: venture capitalists, finance companies, insurance companies, pensions, mutual funds, hedge funds, investment banks, credit rating agencies
- The Government

THE VARIETY OF F.I.s (CONT'D.)

- F.I.s on the periphery
 - ➔ Gambling, pawnbrokers, payday lending, title lenders, loan sharks

DEPOSITORY F.I.s

- Commercial Banks
 - ➔ Center of the financial intermediation universe
 - ➔ Twin roles as distributor of currency and as producer and servicer of demand deposits
 - ➔ Hold a great variety of earning assets
 - ➔ All shareholder owned
 - ➔ Top 5 U.S. banks own > 40% of industry assets. Increasing consolidation.

ASSETS OF TOP 5 US BANKS (2024)

		<u>\$ millions</u>
1	JPMorgan Chase Bank	\$3,510,536
2	Bank of America	\$2,550,584
3	Wells Fargo Bank	\$1,719,839
4	Citibank	\$1,678,936
5	U.S. Bank National Association	\$664,924

TOP 5 US BANKS' ASSETS

- Collectively, the top 5 US banks have assets that exceed the GDP of every country in the world except China and the US!
- In some cases, the largest banks in Europe (HSBC (UK), ING (Netherlands)) have assets that are almost equal to the GDP of the country they are headquartered in, and in some cases even exceed this GDP (Santander (Spain))! How do you “save” these banks if they fail?

DEPOSITORY F.I.s (CONT'D.)

- **Thrifts**

- ➔ More narrowly specialized asset portfolios ($\leq 20\%$ of assets for commercial loans, $\geq 65\%$ in mortgages): residential mortgage loans, consumer savings, marketable securities, etc.
- ➔ Substantially mutual
 - ⇒ Largest S&L was WaMu. In 2008, after Lehman failure, it lost \$16.7 billion in deposits $\Rightarrow$ FDIC took over and sold to JPM Chase for \$1.9 billion.
 - ⇒ Used to be supervised by OTS, but OCC after Dodd-Frank. Most converting to credit unions to avoid (onerous) OCC regulations.

- **Credit Unions (4,742 in 2012)**

- ➔ Specialized in consumer savings
- ➔ Exclusively mutual
- ➔ Supervised by NCUA

NONDEPOSITORY F.I.s (key for start-ups)

Venture capitalists

Finance companies

Insurance companies

Pensions

Mutual funds

Hedge funds

- Actively managed – pursue non-traditional strategies. Private investment pool. Take long and short positions in equities, bonds, currencies.
- No limits on fees (like for mutual funds)
- Unlike mutual funds, no limits on leverage.

Investment banking

Credit Rating Agencies

THE ROLE OF GOVERNMENT

- Vast government enterprises that routinely provide a wide variety of financial services, for example:
 - ➔ Old Age
 - ➔ Survivors and Disability Insurance
 - ➔ Workers' Compensation
 - ➔ Medicare

THE ROLE OF GOVERNMENT (CONT'D.)

- ➡ Housing agencies (Fannie Mae, Freddie Mac, Ginnie Mae): Fannie and Freddie went into conservatorship in September 2008.
- ➡ ...
- Government injected PS in 2008. Amended in 2012 to sweep all profits to U.S. government
- Still in conservatorship but stock of each is traded on NASDAQ
- The U.S. government is far and away the largest F.I. in the country and arguably in the world.
- Fannie Mae (Federal National Mortgage Association): U.S. GSE, publicly-traded company. Founded in 1938 (public since 1968). Securitizes mortgages → creates MBS from conventional mortgages (not insured by FHA, VA, etc.), plus subprimes (since 1995).

THE ROLE OF GOVERNMENT (CONT'D.)

- Ginnie Mae (Government National Mortgage Association): Government owned company that guarantees MBS and created from mortgages guaranteed by FHA and VA. If loans default and FHA or VA don't fully cover investors' losses, Ginnie makes up the difference. Borrows in debt markets by selling bonds and then invests in loans that it securitizes.
- Freddie Mac (Federal Home Loan Mortgage Corp.): Like Fannie, privately-owned GSE. Unlike Ginnie but like Fannie, it goes long in mortgages and MBS. Buys mortgages and securitizes. Created in 1970 to compete with Fannie.

FANNIE, FREDDIE AND US TREASURY

- Fannie and Freddie financially troubled during 2007-09 crisis.
- Congress created Federal Housing Finance Agency (FHFA) in 2008: power of conservatorship over Fannie and Freddie (GSEs). Put GSEs in “temporary conservatorship” soon after.
- Treasury committed to provide \$100 billion in capital via PS purchases (10% annual

FANNIE AND FREDDIE AND FHFA

dividend paid quarterly). Plus \$1 billion commitment fee and warrants to buy 79.9% of common stock at nominal price.

In August 2012 : “ net worth sweep” wiped out all other PS and CS investors.

GSEs have already paid back in dividends substantially more than Treasury put in.

F.I.s ON THE PERIPHERY

Gambling	Difference between gambling and insurance?
Pawnbrokers	Traditional AB (secured) lending. 15,000-20,000 in U.S.
Payday Lending	12,000 by 2004. Provide unsecured ST loans. Borrower writes check to lender for loan or authorizes to debit bank account. Gross margin 30 to 35%; loss 1-1.5% of receivables. Many states have disclosure requirements on effective interest rates, but not limits.
Title Lenders	Secured loans. Hold title to an asset as collateral. Extension of pawn-broking. Interest rates $\geq 25\%$ /month, APR = 300%. Some states limit rates.
Loan Sharks	Unique technology for collections.

INVESTMENT BANKING VIDEO

<https://www.youtube.com/watch?v=9z70BKwfSUA>

As you watch the video, list the key terms you would like to discuss:

CONCLUSION

- Finance is the most global thing we have ... what happens in the financial market affects us all!
- The deposit revolution continues to reshape deposit-dependent F.I.s; depository F.I.s are very likely to be fundamentally restructured in the next 5 years.
- Major competitors for commercial banks and thrifts include insurance companies, finance companies, pensions, and mutual funds.
- One can never forget the role of the government in the financial services industry.

“All essential knowledge relates to existence, or only such knowledge as has an essential relationship to existence is essential knowledge.”

**Soren Kierkegaard:
*Concluding Unscientific Postscript***

Module II:

Economic Services Provided by Banks and the Financial System

**Contemporary Financial Intermediation, Chapter 3.
The What, How, and Why of Financial Intermediation**

KEY QUESTIONS

- Why do we need banks to intermediate between borrowers and lenders?
- Why do we regulate banks and other depository F.I.s so intrusively?
- If banks need to be regulated, how should they be regulated?

ORIGINS AND ECONOMICS OF BANKING

- See Chapters 1 and 2 in Purpose of Banking

INTRODUCTION

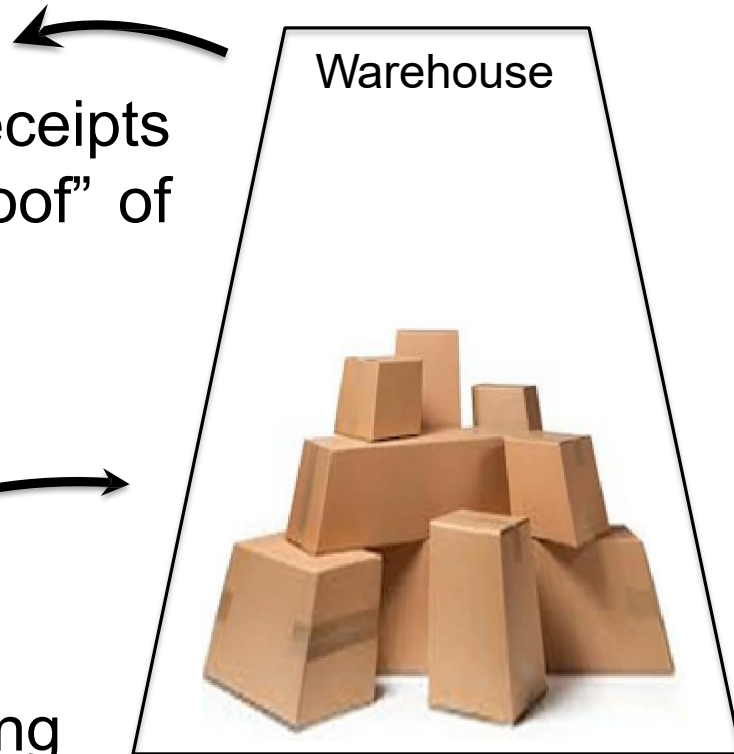
- Which are the second and third oldest professions known to mankind?
 - ➔ Goldsmiths
 - ➔ Pilots
 - ➔ Lawyers
 - ➔ Bankers
 - ➔ Pizza parlor operators
 - ➔ Laundry operators
 - ➔ Circus operators
- And ... at the epicenter of the financial system are banks. What do they do?
 - ➔ Answer: They create liquidity out of thin air!

- Invention of banking preceded the invention of coinage by several thousand years.
- Banking seems to have originated in ancient Mesopotamia and some of the earliest recorded laws pertaining to banks (banking regulation) were part of the Code of Hammurabi.
- Deposits: cattle, grain, precious metals
 - Loans made
 - Interest paid
- In ancient Egypt, grain harvests were “deposited” (or stored) in centralized state warehouses...depositors could write written orders for the withdrawal of a certain quantity of grain as a means of payment.

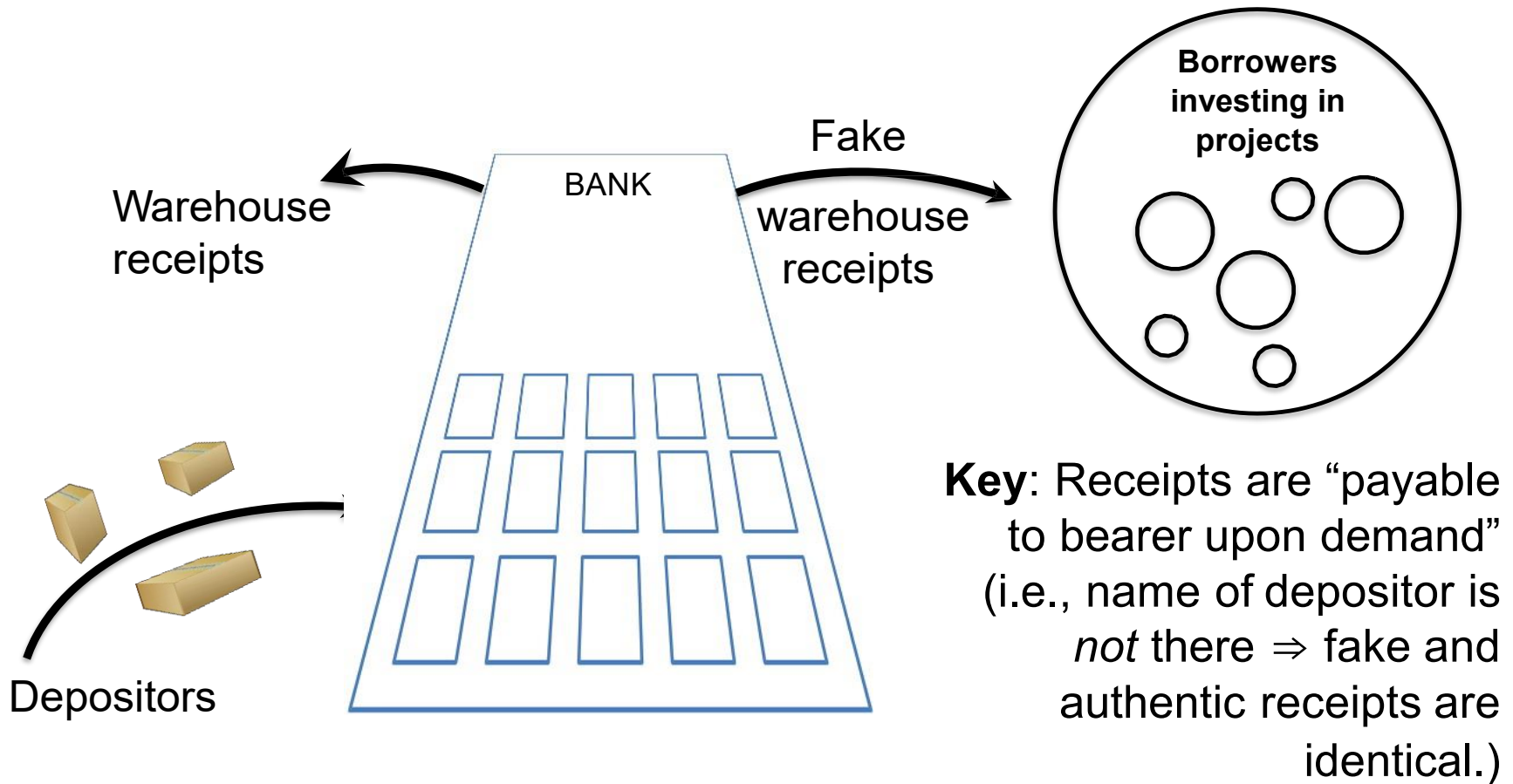
Basic Idea

Warehouse receipts
issued as “proof” of
deposit

Depositors bring
grain or gold bricks
to warehouse.



Basic Idea



MAIN RESULTS

- In order to create liquidity banks must play two roles:
 1. deposit-taking and warehousing
 2. lending
- Critical for lending will be the creation of private money, that we will call “fake receipts”
- Fake receipts are receipts not backed by deposits... and key is these are indistinguishable from “authentic receipts”
- In the existing literature, the bank takes in deposits and lends them out. That is, deposits create loans. What we show is that loans create deposits!

Our Approach Is Reminiscent Of Keynes...

“It is not unnatural to think of deposits of a bank as being created by the public through the deposits of cash representing either savings or amounts which are not for the time being required to meet expenditures. But the bulk of the deposits arise out of the actions of the banks themselves, for by granting loans, allowing money to be drawn on an overdraft or purchasing securities, a bank creates a credit in its books which is the equivalent of a deposit.”

(Keynes in his contribution to the
Macmillan Committee, 1931, p. 34)

IN OUR MODEL

- Liquidity is created only via banks
- Firms can invest over and above depositors' endowments
 - ➔ because banks' fake receipts provide working capital to firms
- Swiss Referendum: what % of deposits were fake receipts?

Over the past 20 years, on average “fake receipts” (private money creation by banks) accounted for about 90% of all U.S. bank deposits ... so this theory applies even more to contemporary banks than to ancient banks.

FIAT MONEY, FEDERAL RESERVE AND U.S. TREASURY

- Why does the government need to borrow when it can print its own money?

FCM & MONETARY POLICY

- 12 Federal Reserve Banks, each with its own Board of Directors: Boston, New York, Philly, Cleveland, Richmond, Atlanta, Chicago, St Louis, Kansas City, Minneapolis, Dallas, and San Francisco. Presidents report to their own boards and the Board of Governors of the Federal Reserve System.
- Open market operations.
 - ➔ Purchases of government securities (Treasuries) → increase in bank lending as purchases put more liquidity in the hands of banks.
 - ➔ Sales of government securities has the opposite effect, as liquidity is taken out of the banking system.
- Reserve requirement changes.
 - ➔ Increase in reserve requirements → less lending.
 - ➔ Reduction in reserve requirements → more lending.
- Discount rate changes.
 - ➔ Higher discount rate → less lending.
 - ➔ Lower discount rate → more lending.

REFLECTION

- Take 15 minutes in your groups to discuss:
- (1) Three key takeaways so far
- (2) Remaining Questions

“Robert M. Greer is apartment hunting, even though he doesn’t need a place to live. What the Lones Lang Wooton managing director is seeking is the best apartment building for inclusion in a securitized mortgage portfolio.”

American Banker, October 2, 1990

Module III

Securitization

Contemporary Financial Intermediation Chapter 11

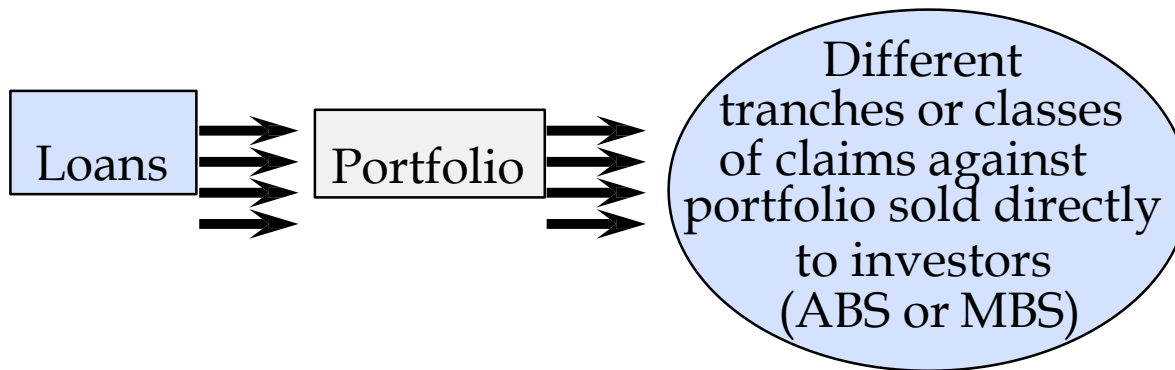
Securitization

KEY QUESTIONS

- What is securitization? How does it differ from loan sales?
- What assets are securitized? What is the economic motivation for securitization? What are the different types of securitization contracts?
- Why do banks securitize assets? What does securitization do for both the originator and the investors?
- Can and should everything be securitized? How much is right?
- How will securitization affect the evolution of the banking industry?

What Is Securitization? How Does It Differ From Loan Sales?

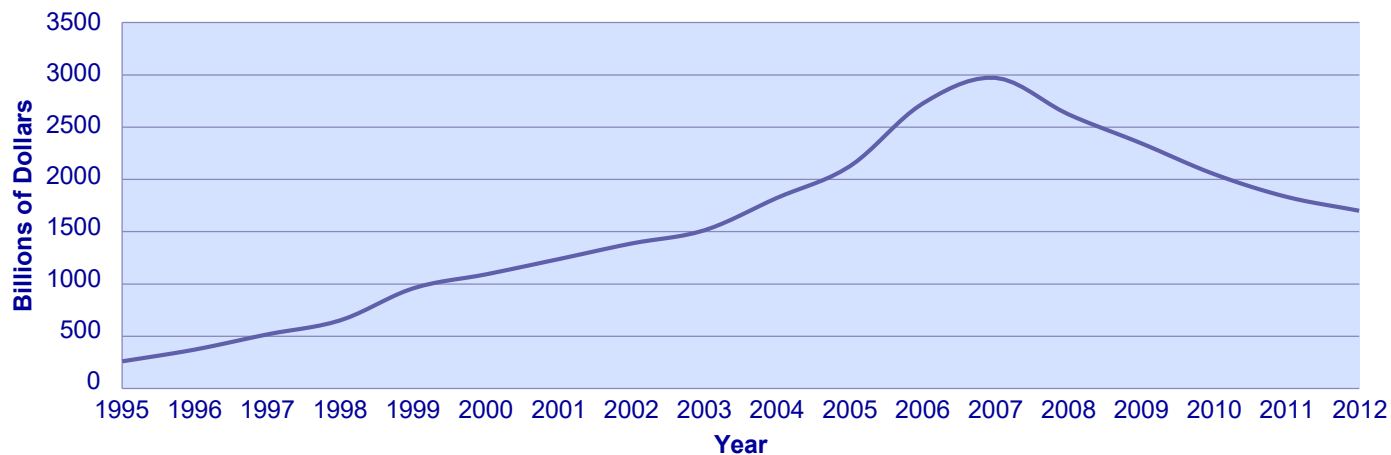
- Securitization involves combining the loan with others of similar characteristics, creating credit-enhanced claims against the cash flows of this portfolio, and then selling these claims to investors



⇒ Increases options available to investors

How Large Is The Securitization Market?

- The securitization market, known as the market for Asset-Backed Securities (ABS), grew to almost \$2 trillion by 2007:

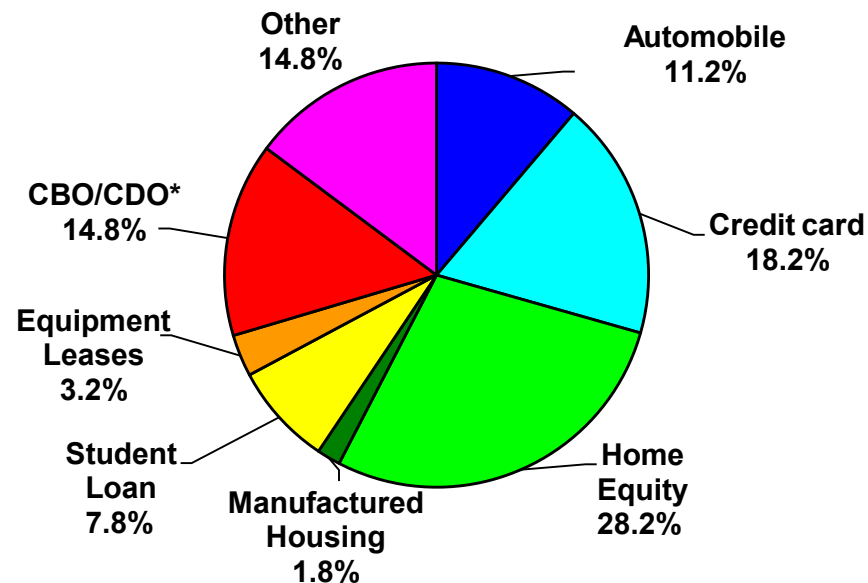


- Private issuers include commercial banks, financial subsidiaries of industrial companies, and savings institutions

WHAT ASSETS ARE SECURITIZED?

- Currently, a wide range of assets are securitized:
 - ➔ Examples are automobile loans and leases, credit card receivables, commercial truck loans, and boat loans

ABS Outstanding
by Major Types of
Credit in 2005



* CBO is Collateralized Bond Obligation; CDO is Collateralized Debt Obligation.

What Is The Economic Motivation For Securitization?

- Decomposition of lending into four basic operations: origination, guaranteeing, servicing, and funding
- Why are these functions combined in the first place but decomposed now?
 - ➡ Funding advantages due to the regulatory environment and information technology

What Is The Economic Motivation For Securitization? (Cont'd.)

- The funding advantage in the earlier times was a consequence of regulation:
 - ➔ Deposit interest rate ceilings, underpriced governmental deposit insurance, entry restrictions, and various tax advantages
 - ➔ Implicit contract between depositors, banks/thrifts, and the government
- Incentives for securitization and loan sales:
 - ➔ Dismantling of deposit interest rate ceilings → erosion of funding benefits
 - ➔ Advances in information technology (Example 9.1)

What Are The Different Types Of Securitization Contracts?

- Most common: pass-throughs
 - ➔ Investors have direct ownership in a portfolio of mortgage loans that share similar maturity, interest rate, and quality characteristics
 - ➔ Ownership of the loan rests with certificate holders (investors); thus, pass-throughs do not appear on the originator's balance sheet

What Are The Different Types Of Securitization Contracts? (Cont'd.)

➡ Two structures:

❖ Static pool pass-through

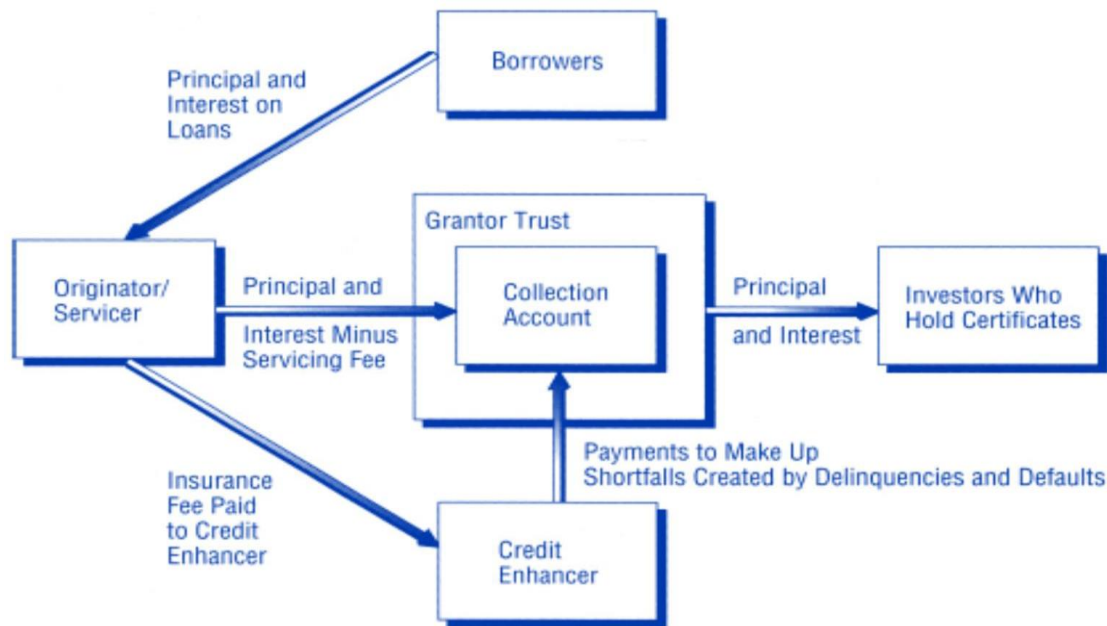
- “Static” means that the pool of loans against which claims are sold to investors is fixed
- Example: mortgage-backed security

❖ Dynamic pool pass-through

- “Dynamic” means that the debt obligations included in the pool are usually short term, so that they turn over, implying changes in the composition of the loan portfolio
- Most often used with credit card receivables

Different Types Of Securitization Contracts

- Pass-through
 - ➡ Cash-flow schematic for a static pass-through:
- Trust sometimes called Special Investment Vehicle (SIV) or Special Purpose Corporation (SPC)



Securitization Innovations

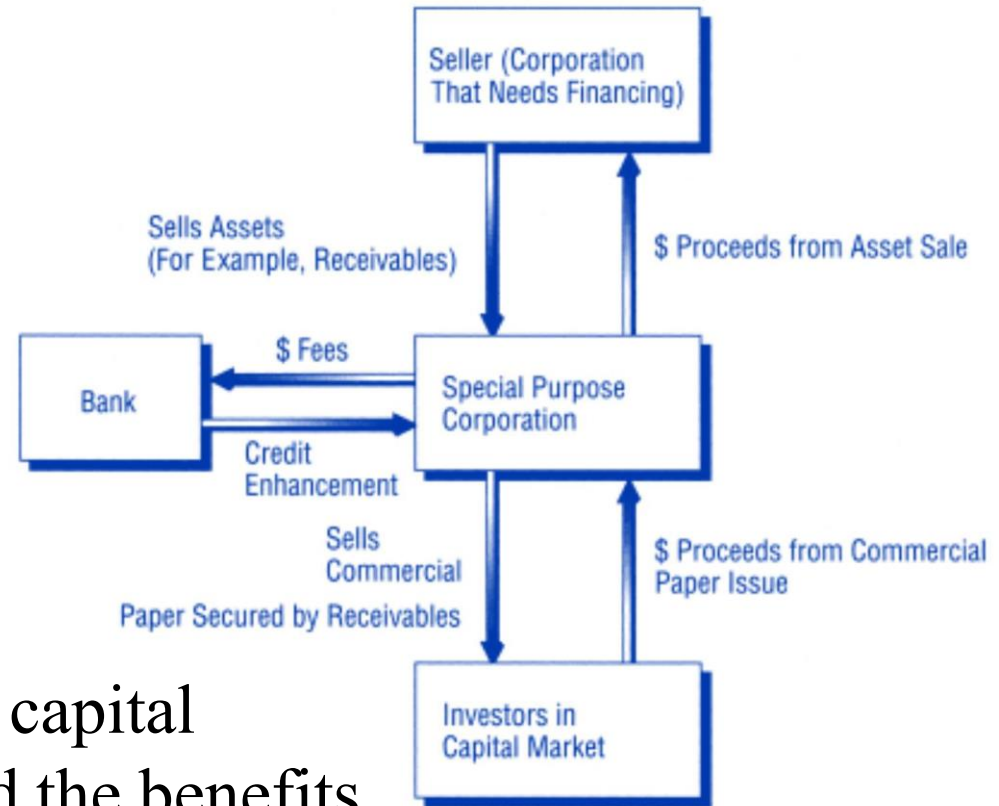
- Asset-Backed Commercial Paper (ABCP)
 - ➔ Commercial paper secured by designated corporate assets, typically receivables
 - ➔ Oxymoronic since commercial paper is defined as short-term, unsecured debt obligation of a corporation
 - ➔ Market came into existence in 1983 and has expanded to a multibillion-dollar market
 - ➔ Offers some firms lower cost of funding than regular (unsecured) commercial paper or a bank loan

SECURITIZATION Innovations

- Asset-Backed Commercial Paper (ABCP)

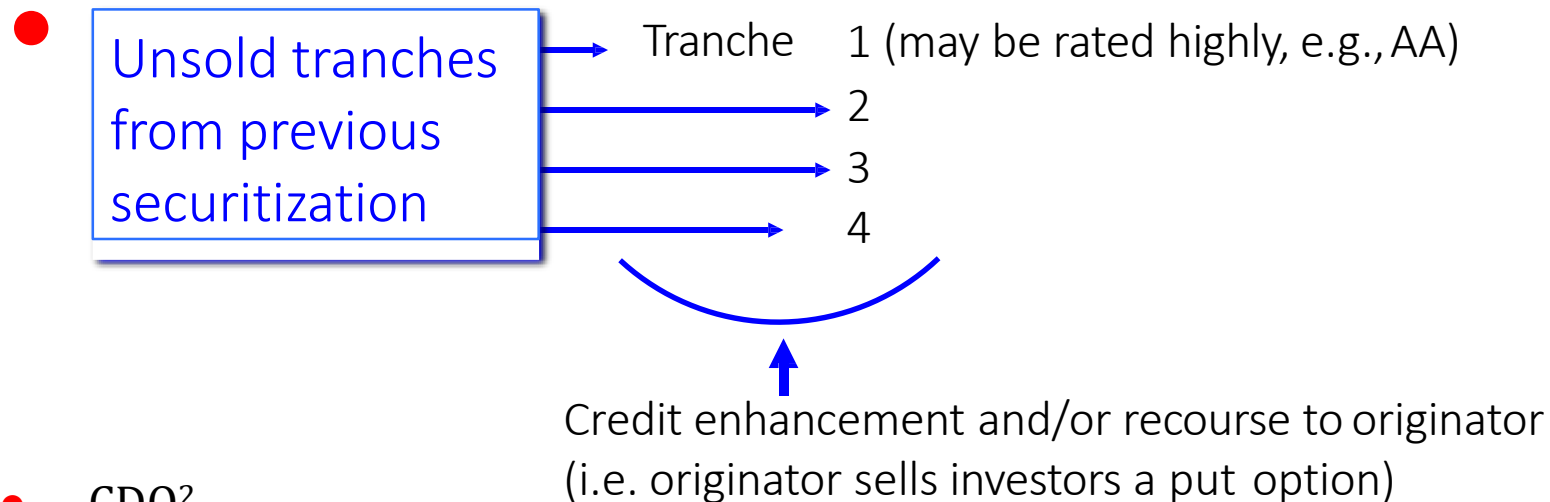
➔ A typical ABCP program:

➔ The risk-based BIS capital rules have increased the benefits of ABCP to banks (not posting as much capital as with conventional funding)



CDOs and CDS

- Since 2009, related structure call Re-REMICs – static collateral trusts – was issued. Many CDOs are dynamic in that collateral is changed over time.
- Collateralized debt obligations (mainly with mortgages)



- CDO²
- CDO market has shrunk since crisis
- Also CLOs (Collateralized Loan Obligations)
- CDS: how is it different from default insurance?

WHAT IS A PUT OPTION?

- I buy *option*
to sell a security
at a future date T
at a fixed “strike price” S .

- My payoff at exercise:

$$= \begin{cases} S - P_T & \text{if } S > P_T \text{ where } P_T \text{ is price of security at date } T \\ 0 & \text{otherwise} \end{cases}$$

- Value of option increases as volatility (risk) of P_T increases.

⇒ Recourse is more valuable for riskier pools.

Securitization of Other Assets

- Auto loans
 - ➔ *Certificates of Automobile Receivables* or “CARS”
 - ➔ Securitization of retail installment sales contracts that are backed by autos and light trucks
 - ➔ Usually pass-through securities involving a higher servicing fee than mortgage-backed securities

Securitization Of Other Assets (cont'd.)

- Credit cards
 - ➔ Securitization of credit-card receivables: Certificates of Amortization Revolving Debts or “CARDS”
- Other tangible assets
 - ➔ Computer leases, trade credit, financial leases, junk bonds, leverage buyouts, commercial loans, solar panels, song royalties, proceeds from tobacco litigation, anticipated lawsuit proceeds, loans for manufactured homes, unsold airline seats.
- Intangibles like intellectual property
 - ➔ Trademarks, brand names, logos, patents, etc.

Why Securitize?

- **Benefits to the issuer (the supply side):**

1. Management of interest rate risk

- ❖ Decreases maturity mismatch
- ❖ Reduces prepayment risk
- ❖ Lengthens the average maturity of liabilities

2. Increased liquidity

- ❖ Trading increases information available about the asset in the public domain; high-priority tranches are least information-sensitive and most liquid.
- ❖ Third-party credit enhancement reduces the effect of informational asymmetries between the issuer and investors

Why Securitize? (Cont'd.)

- **Benefits to the issuer (the supply side):**

- 3. Diversification of funding sources

- ❖ Provides a way of raising financing from sources more than traditional ones
 - ❖ Thus, better diversification of funding sources

WHY DO BANKS SECURITIZE ASSETS? WHAT DOES SECURITIZATION DO FOR ISSUERS?

- **Benefits to the issuer (the supply side):**
 - 4. Avoidance of intermediation taxes
 - ❖ Help avoid reserve and capital requirements and deposit insurance premiums

WHY SECURITIZE? WHAT DOES SECURITIZATION DO FOR INVESTORS?

- **Benefits to investors (the demand side):**
 1. Provides a greater variety of cash-flow streams that investors can hold ➡ more options to invest:
 - i. Claims produced were not previously available to investors through linear combination of existing claims
 - ii. Provides a less-expensive alternative in terms of transaction costs
 - iii. Allows institutions like insurance companies to get access to very long-maturity assets and banks to have access to very short-maturity assets
 - iv. Hedge funds like lower tranches with higher yields

WHY SECURITIZE? WHAT DOES SECURITIZATION DO FOR INVESTORS?

- **Benefits to investors (the demand side):**

- 2. Liquidity

- ❖ Assured by the size of the asset-backed securities market and the active trading involved

Strategic Issues In Securitization For Banks As Lenders

A. Securitization as a balance sheet management tool

- ➔ F.I.s must carefully balance the gains from laying off risks through securitization against its cost
- ➔ Balance sheet funding will be chosen when: FI has funding cost advantage to make it profitable to hold asset on B/S
- ➔ Securitization helps to diversify the assets on the balance sheet

B. Securitization as a pricing tool

- ➔ Can help improve the accuracy of bank's pricing
- ➔ Also, permits the bank to either lower its prices to increase market share or earn higher profit margins at current prices

Strategic Issues in Securitization (cont'd.)

- C. Securitization as a strategic weapon for market penetration and diversification
- ➔ Ensures the bank greater continuity of presence in its strategically chosen market segment
 - ➔ Permits banks to overcome geographical and industrial limitations and hence facilitates diversification

What is Shadow Banking?

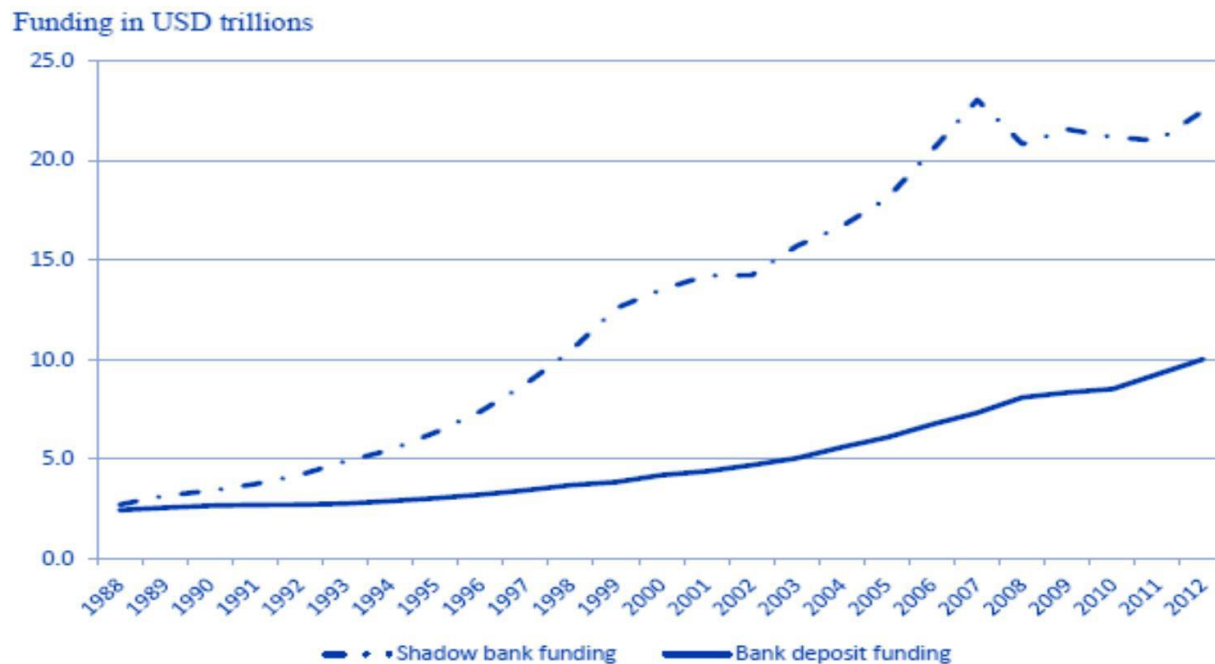
- In the U.S., the 2007–09 subprime crisis started in the shadow banking system.
- Shadow banks are institutions that engage in credit and maturity transformation, just like banks, but without deposit funding.
- Shadow banks include a variety of non-depository institutions like investment banks, insurance companies, mutual funds, hedge funds, and finance companies; securitization structures such as ABCP and ABS; and key investors in securitized structures, such as money market mutual funds.

How do Shadow Banks fund themselves without deposits?

- With “repurchase agreements” or “repos”
 - ➔ With a repo, an institution borrows short-term using traded securities (like Treasury bonds) as collateral, having the collateral returned to it when the loan is repaid.
 - ➔ The collateral used in these transactions comes from the asset-backed securities created by the securitization of loans, leases and mortgages.

How do Shadow Banks fund themselves without deposits?

- Shadow banking has now *eclipsed* traditional depository funding in the U.S.



Role of Commercial Banks in Shadow Banking

- In an increasingly interconnected financial services universe, shadow banks and commercial banks are joined at the hip...
- Most obvious connection is that a bank holding company (BHC) may own both a commercial bank and a shadow bank...
- Or a BHC may have an asset-backed commercial paper (ABCP) conduit, which may be off-balance-sheet to the BHC, but may be supported by a commercial banking subsidiary of the BHC through loan commitments.
- Also, some of the collateral used in repo transactions comes from the securitization of loans made by commercial banks.

CONCLUSION

- Securitization involves qualitative asset transformation of loans into traded securities.
- Securitization may be the most important engine of reform in our financial system.
- The trend toward securitization is likely to continue, as long as solutions to the moral hazard problem can be developed.
- Securitization is a key part of shadow banking, which is now larger than depository banking.

GROUP EXERCISE #1: CASE STUDIES IN FINANCIAL ENGINEERING

- Consider the three short caselettes (tales). Brainstorm a creative solution to each problem.
- List your takeaways from this exercise about how financial engineering can help you tap the power of global financial markets to create new business ideas.

Case 1:

The first tale is about the December 2015 Paris Climate Accord. Major enthusiasm followed because there was widespread agreement on global emission reductions and pledges to increase investments in renewable energy. This was welcome news to supporters of clean energy in light of diminishing enthusiasm on the part of traditional venture capitalists to invest in renewable-energy companies. However, not long after the Paris Accord, there was bad news. SunEdison filed for bankruptcy in April 2016. The company was planning to become a big player in renewable energy. But its bankruptcy cast a pall over the prospects of private-sector initiatives to promote clean, renewable energy. What should be done to regain momentum?

Case 2:

For the second tale, imagine that you own a house in an area in which there is plenty of sunshine most of the year, and you spend a fortune on air conditioning, which you also need most of the year. Along comes a company that wants to sell you solar panels for the house so you can convert the whole house to solar energy. You like the idea because you care about the environment. The problem is the cost. The estimate is \$100,000. You just can't afford that right now. What do you do?

Case 3:

For the third tale, imagine a somewhat different challenge. There is a community with a large inventory of unsold houses—some of these are houses that were foreclosed on during the Great Recession, and others are houses the homeowners want to sell but are unable to get a good price for. There are just not enough homebuyers. In the community there is ironically also a shortage of rental housing. What should be done?

GROUP EXERCISE #2:

CASE STUDY: LONE STAR BANK

(From Chapter 11 in the *Contemporary Financial Intermediation* book)

- Question: Address the question under “The Assignment” (p. 280) at the end of the case (*not* the review questions at the end of the chapter).

“In its analysis of the crisis, my testimony before the Financial Crisis Inquiry Commission drew the distinction between triggers and vulnerabilities. The *triggers* of the crisis were the particular events or factors that touched off the events of 2007-09 – the proximate causes, if you will. Developments in the market for subprime mortgages were a prominent example of a trigger of the crisis. In contrast, the *vulnerabilities* were the structural, and more fundamental, weaknesses in the financial system and in regulation and supervision that served to propagate and amplify the initial shocks”.

Chairman Ben Bernanke (April 13, 2012).

Module IV:

The Financial Crisis

Contemporary Financial Intermediation, Chapter 14.

The 2007–09 Financial Crisis and Other Financial Crises

The Purpose of Banking, Chapter 8.

Key Questions

- What were the events that led up to the financial crisis of 2007-09?
- What were the causes of the crisis?
- What were the effects of the crisis on the real sector?
- What were the policy responses to the crisis? Bailouts? What should they have been?
- What are the most important lessons we learned from this crisis?

WHO TO BAIL OUT?

- Bailouts sometimes unavoidable to prevent a meltdown, but they cost taxpayers money and create moral hazard.
- So have to be selective. Who to bail out?
- [South Park - American Economics - YouTube](#)

What Were the Events that led up to the Financial Crisis Of 2007–09?

- The financial crisis of 2007–09 is widely regarded as the worst since the Great Depression of the 1930s.
- Crisis rooted in housing boom that occurred in the U.S. prior to 2006.
- Credit in the U.S. began to tighten in 2006, causing house prices to begin to decline.
- In early 2007, Freddie Mac announced it would no longer purchase high-risk mortgages + New Century Fin. Corp. filed for bankruptcy

What Were the Events that led up to the Financial Crisis Of 2007–09?

- Heightened solvency concerns
- Large-scale withdrawals in Aug. 2007 from various markets.
- Increased “haircuts” on repos that substantially contracted short-term borrowing capacity in shadow banking in the U.S.

What Were the Events that led up to the Financial Crisis Of 2007–09?

- U.S. Federal Reserve opened up short-term liquidity provision facilities for various institutions

But...

Problems continued and things even got worse.

What Were the Events that led up to the Financial Crisis Of 2007–09?

- Early 2008: stresses resulted in institutional failures:
 - ➔ Countrywide financial (mortgage lender) bought by BOA (Jan. 2008)
 - ➔ J.P. Morgan Chase bought failing Bear Stearns (Mar. 2008)
 - ➔ Indy Mac (largest mortgage lender in the U.S.) was taken over by the U.S. government
 - ➔ Fannie Mae and Freddie Mac became financially distressed and were taken over by the government (Sep. 2008)

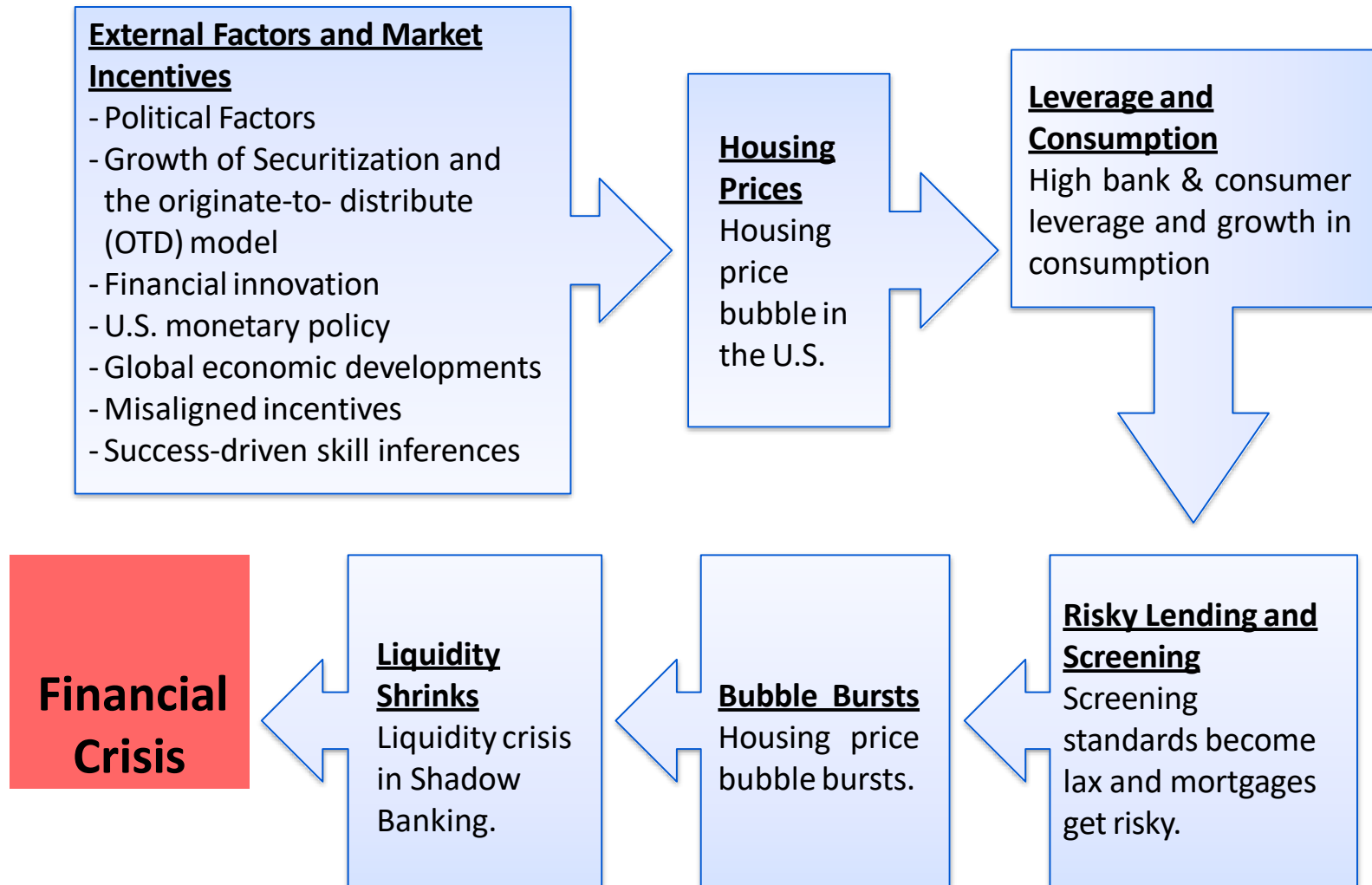
What Were the Events that led up to the Financial Crisis Of 2007-09??

- Sep. 15, 2008:
 - ➔ Lehman Brothers filed for Chapter 11 bankruptcy.
 - ➔ AIG, a leading player in CDs market, received \$85 billion in government assistance.
- Sep. 16, 2008:
 - ➔ Reserve Primary Fund, a money market fund, “broke the back” (per share value fell below \$1)

What Were the Events that led up to the Financial Crisis Of 2007–09?

- Sep. 25, 2008:
 - ➔ A large S&L (Washington Mutual) taken over by FDIC and assets transferred to J.P. Morgan Chase.
- October 2008:
 - ➔ Crisis Spreads to Europe
- 2009:
 - ➔ Crisis continues. U.S. unemployment rate hits 10%

What Were the Causes of the Crisis?



External Factors and Market Incentives:

- *Political Factors:*

- ➡ Legislative initiatives to broaden home ownership
➡ growth in subprime mortgages ➡ housing price bubble and indebtedness by consumers who could barely afford their homes.
- ➡ Deregulation and regulatory capture.

External Factors and Market Incentives:

- *Growth of Securitization and OTD model:*
 - ➔ Easy money monetary policy;
 - ➔ softer lending standards by banks;
 - ➔ diluted screening by banks due to OTD model;
 - ➔ increase in credit supply and credit risk

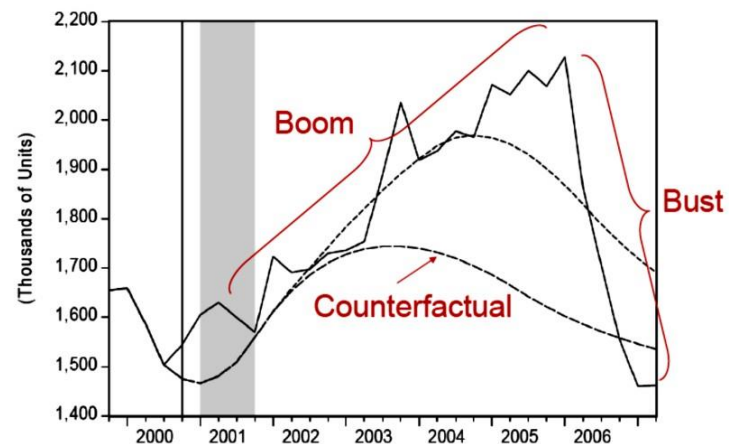
External Factors and Market Incentives:

- *Financial Innovation:*

- ➔ New instruments to raise financing that exposed financial institutions to greater rollover risk ...
e.g., Credit Default Swaps (CDS)

- *U.S. Monetary Policy:*

- ➔ Easy money with unusually low interest rates ➡ inflation in house prices ➡ House price bubble and excessive building of houses.



External Factors and Market Incentives:

- Same situation in Europe:
- The European country with the largest deviation from the Taylor rule had the biggest housing boom (Spain)
- *John Taylor*: as inflation increases, how should the central bank adjust nominal interest rate?

Target short-term nominal interest rate (e.g., fed funds rate)

$$= i_t = \pi_t + r_t^* + \alpha_\pi (\pi_t - \pi_t^*) + \alpha_y [y_t - \bar{y}_t]$$

Rate of inflation $\uparrow$ $\downarrow$ $\downarrow$ $\uparrow$ $\uparrow$ $\swarrow$ $\nwarrow$
 Equilibrium real interest rate (≈ 0.5) Desired inflation rate (≈ 0.5) Log (real GDP) Log (potential GDP)



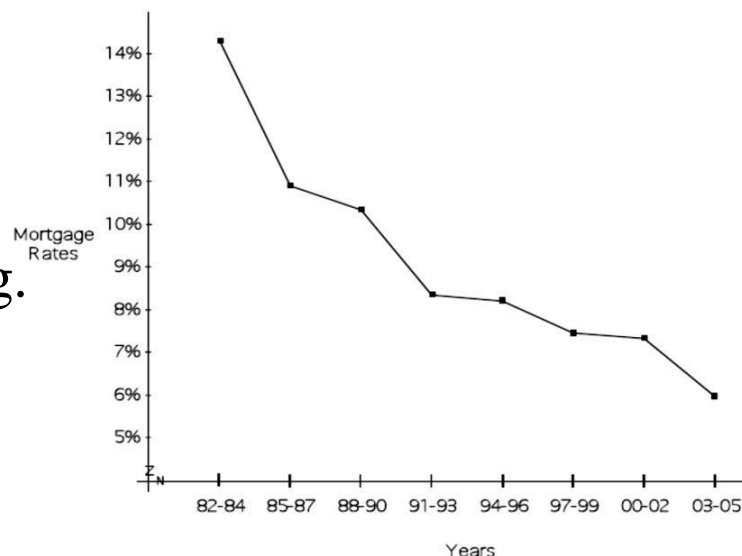
- As inflation $\uparrow$ (relative to target), i_t is increased (same for GDP) to cool down economy.
As inflation $\downarrow$, i_t is decreased to stimulate growth.

EXTERNAL FACTORS AND MARKET INCENTIVES:

- *Global Economic Developments:*

- ➔ Emergence of countries like India, China and Brazil generated wealth in those countries.

- ❖ They invested their money in the U.S., flooding the U.S. with liquidity, which went into housing and kept mortgage interest rates falling.



External Factors and Market Incentives:

- Misaligned Incentives:

- ➔ Too-Big-to-Fail safety nets and deposit insurance.
 - ❖ excessive risk-taking by financial institutions

- Success-Driven Skill Inferences:

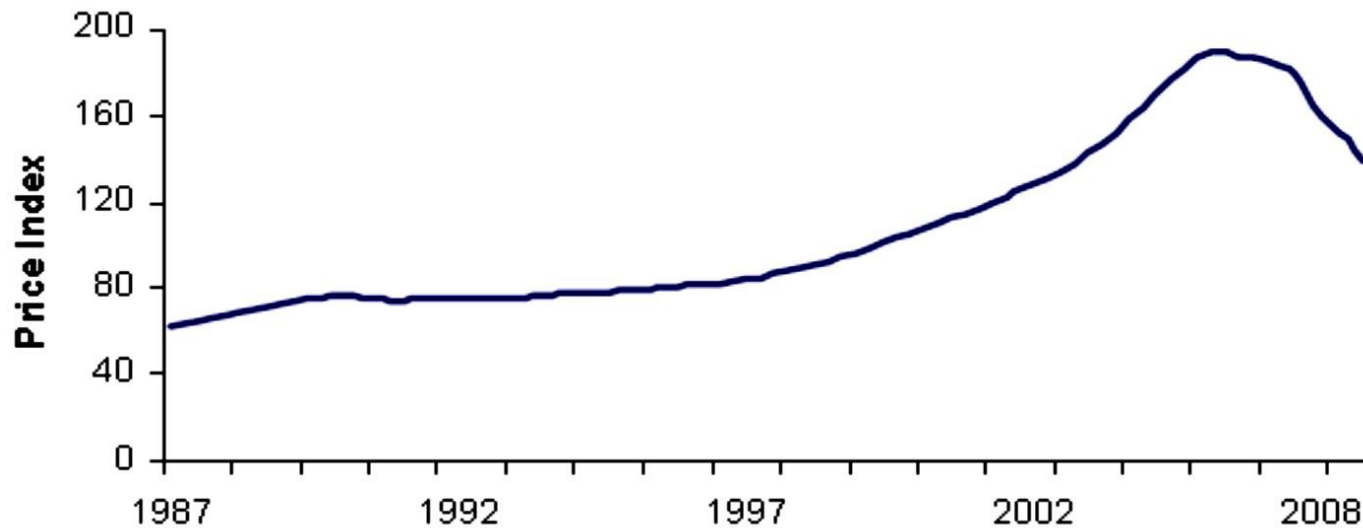
- ➔ Long period of tranquility and bull market between 1995 and 2005.
 - ❖ elevated perceptions of skills/abilities of bankers.
 - ❖ Investors willing to fund banks that invest in riskier (new) products (that are more profitable for banks)

External Factors and Market Incentives:

- ❖ Eventually, if external events lead investors to revise their beliefs that outcomes are due to bankers' skills and believe instead that they are due to luck...
- ❖ key to theory is that it explains near inevitability of financial crises when things have gone well for a long time.

Housing Prices

- ❖ As a result of factors above, house prices in U.S. experienced big increase prior to crisis...



Leverage and Consumption

- U.S. household consumption grew at the rate of \$1,994 per year during 1980–99, but then jumped to \$2,849 per year from 2001–07.
 - ➔ Higher consumption financed with higher borrowing, which was supported by rising home prices.
- As value of residential real estate, mortgage borrowing increased even faster.
 - ➔ Home equity fell from 58% of home value in 1995 to 52% of home value by 2007.

Leverage and Consumption

- During 2002–05, subprime borrowers experienced an unprecedented growth in credit despite declining relative income growth.
 - ➔ Borrowers who could not afford to borrow more did borrow more due to higher home prices.

Risky Lending and Screening

- Due to securitization (OTD model), banks had weaker screening incentives.
 - ➔ Quality (performance) of loans deteriorated for 6 years prior to the crisis.
- Banks seemed to be aware, at least to some extent, that they were making riskier loans.
- Effect (riskier lending) strongest for banks with low capital.

Bubble Bursts

- As subprime borrowers defaulted, credit rating agencies began to downgrade mortgage-backed securities ➡ Downward spiral set, banks cut back on lending ➡ Adverse impact on house prices.

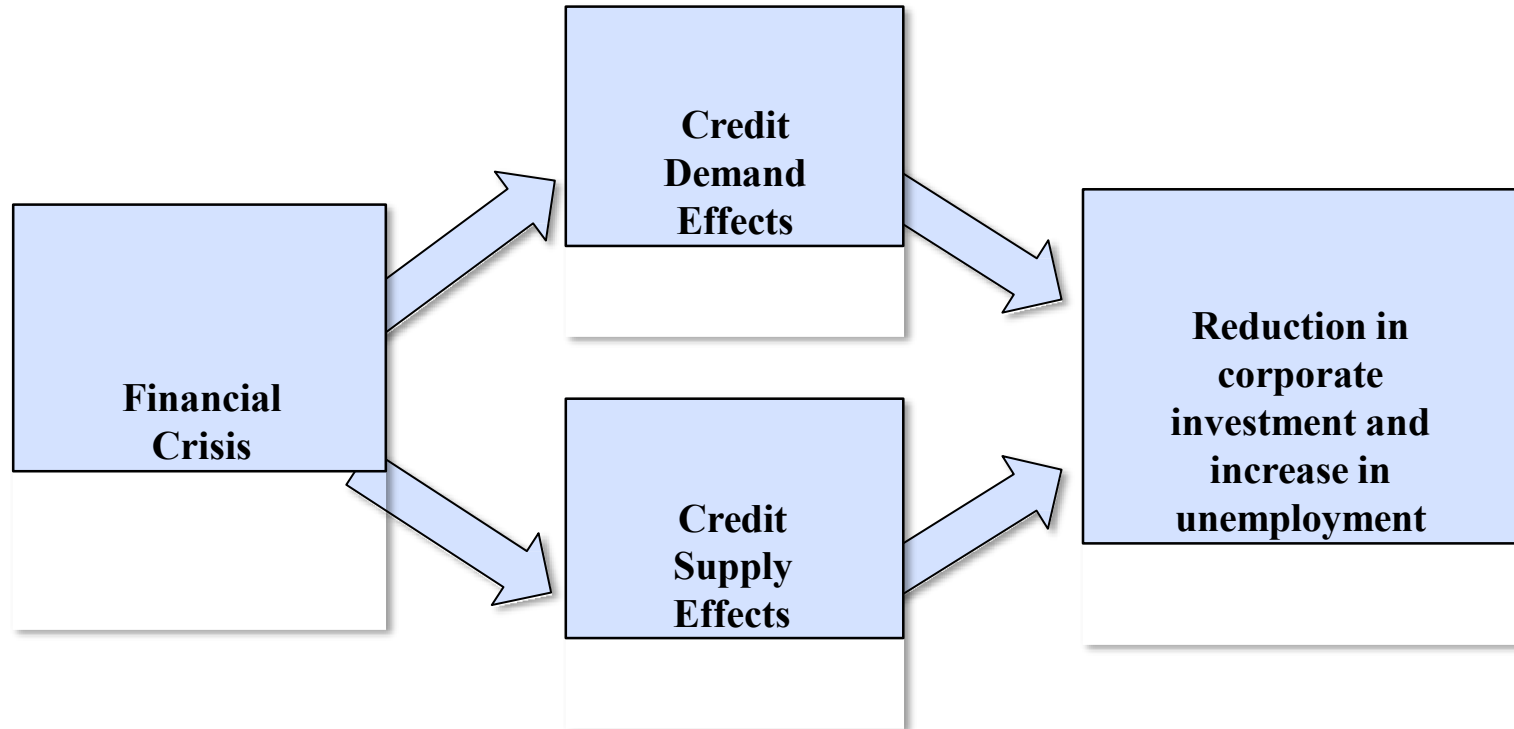
Liquidity Shrinks in Response to Insolvency Concerns

- Shadow banking grew rapidly in the U.S. and Europe prior to the crisis
- Repos common way of doing short-term funding in shadow banking
- Repos used MBS (mortgage-backed securities) as collateral
- Decline in home prices and consequent mortgage defaults
 - ➔ Falling MBS prices
 - ➔ Lower collateral value for repo borrowing

Liquidity Shrinks in Response to Insolvency Concerns

- ➡ Shrinkage of liquidity to financial institutions in shadow banking sector
- Research indicates that this drying up of liquidity was a bank-specific solvency issue, rather than a market-wide liquidity crunch.

What were the Real-Effects of the Crisis?



What were the Real-Effects of the Crisis?

- Credit Demand Effects:

- ➡ Bursting of house price bubble

- ❖ Declining consumer net worth for highly levered borrowers who owned homes $\Rightarrow$ consumers stopped spending money
 - ❖ Collapse of consumption + Demand for housing fell
 - ❖ Further decline in house prices

- Credit Supply Effects:

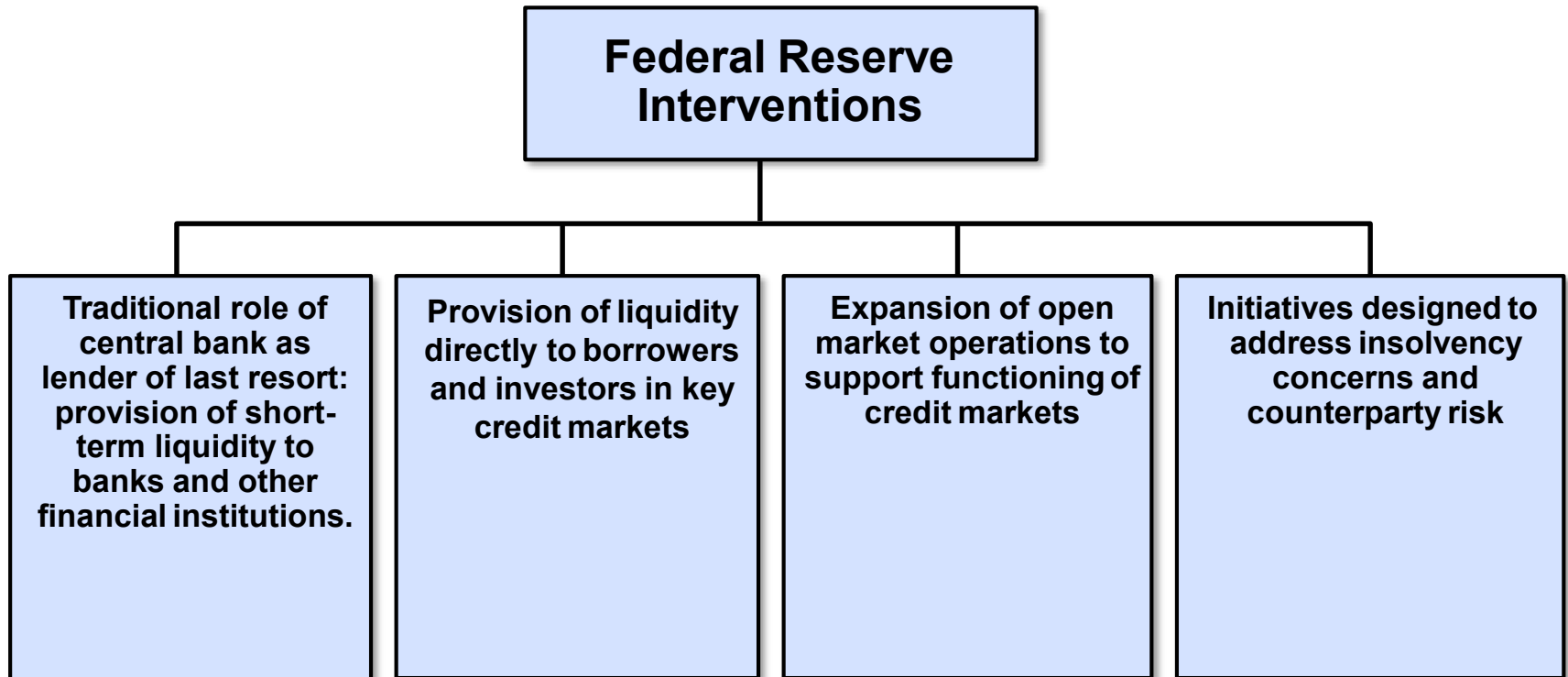
- ➡ Banks took hits to capital due to decline in MBS prices.

- ❖ Reduced credit supply + Increase in price of credit (higher loan spreads)

What were the Real-Effects of the Crisis?

- Reduction in Corporate Investment and Increase in Unemployment:
 - ➔ Due to declining household consumption and scarcer and more expensive credit.
 - ❖ Corporate investment fell and unemployment rose $\Rightarrow$ led to further decline in consumer spending.
 - ❖ U.S. entered deep recession.
 - ❖ Loss of jobs and plunging labor participation rate.
- Impact on GDP:
 - ➔ The Federal Reserve Bank of Dallas estimates that the financial crisis cost the U.S. \$6 trillion - \$14 trillion (\$50,000 to \$120,000) for every U.S. household.

What were the Policy Responses to the Crisis?



What were the Policy Responses to the Crisis?

Bottom Line:

- Initially the Federal Reserve viewed this as a liquidity crisis, and created liquidity-provision facilities (with sunset provisions), but...realized later during the crisis, that this was an insolvency crisis
- Later initiatives were aimed at recapitalizing U.S. banks via equity infusions, some by the government that received equity stakes.

Financial Crises in other Countries

Country	Policy Responses
Sweden	- Nationalization of banks - Loan guarantees - Government ownership
Latvia	- Some banks liquidated - New banking laws passed - Regulatory powers of central bank strengthened - Deposit insurance established
Argentina	- Government defaulted on its debt - Country exited the currency board regime - Restricted deposit withdrawals - Conversion of dollar-denominated loans and deposits into pesos - Regulatory forbearance - Banks closures - Bank nationalizations

What Macroeconomic lessons do We Learn from the 2007–09 Crisis?

- Excessive consumer and bank leverage can lead to unsustainable price bubbles in real estate and other assets, and financial crisis can occur when the bubble bursts
 - ➡ Inject higher capital into banking (via higher capital requirements) and limit consumer leverage through minimum equity requirements
- Policymakers need to determine in real time whether it is a liquidity or insolvency crisis—a difficult task—in order to come up with the right policy initiatives to deal with the crisis.

What are the Macroeconomic Lessons from Financial Crises?

- Evidence indicates that almost every financial crisis has two common pre-crisis features:
 - ➔ High leverage (usually in banks but sometimes also involving households)
 - ➔ Some price bubble in some real assets
 - ➔ Either we are not learning enough from previous crises or we are willing to accept a positive probability of a financial crisis in exchange for the pre-crisis economic growth that often accompanies high leverage and price bubbles.

Conclusion:

- The 2007–09 crisis was the worst since the Great Depression of the 1930s.
- Crisis affected Europe as well. What started in the subprime mortgage market in the US engulfed Main Street and the rest of the world.
- Crisis was caused by excessive leverage and house price bubble that burst when subprime borrowers began to default.
- This caused both credit demand and supply to fall and caused liquidity to shrink in response to insolvency concerns.
- Of the various central bank policy initiatives, the ones that were the most effective were the ones that addressed insolvency concerns (e.g. bank recapitalization).

Organizational Risk Management Lessons Learned from the 2007- 09 Crisis

- What lessons about risk management do we learn from the crisis?
- How would you use those lessons in managing risks in your own organizations?

Lesson #1: Corporate Culture has a Huge Impact on Risk Management Effectiveness

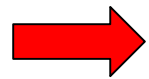
- Culture determines:
 - ➔ How much info sharing is there?
(Jack Welch, Todd B. Story)
 - ➔ Are people willing to be the bearers of ill tidings?
- Lack of cultural compatibility → Daimler-Chrysler, Citicorp-Travelers.
- Culture now a big issue in thinking of bank regulations.
- How do you “measure” culture?

Lesson #2: Market Conditions have a Huge Impact on Risk Appetite and Risk Management Effectiveness

- Financial crisis came about because we had over a decade of sustained growth and high banking profitability...So, everybody – bank managers, regulators, rating agencies, investors – raise their assessments of the skills of bankers in managing risks.

 Four Consequences:

- We become blind to the risks that have not been Recently realized.
- Price creditors demand as compensation for risk falls → This lowers risk premia and makes risk taking more attractive.
- Tends to marginalize risk managers within the firm...viewed mainly as naysayers who want to slow down profit growth!
- Makes having high capital look unattractive because it lowers ROE...but what's the benefit?



IN THE SHORT RUN, GOOD
MARKET CONDITIONS
MAKE RISK MANAGEMENT
EASIER...BUT

- In the long run, there is no bigger enemy of effective risk management than good market conditions!
- Because...risk management gets corrupted when times are good!

Lesson #3: Beware Of Good Times

The seeds of a crisis are sown in the good times (think of the boom during 2002–06 or the ag boom before the 1985 farm credit crisis)

- We need to be vigilant during our time of success. Investment banks over-extended themselves in risky investments due to very high collateral values, and then found that short-term financing dried up.
- How much of our thinking about the future influenced by what has happened in the recent past?

Lesson #4: Keep Enough Equity Capital

- Capital is essential. Keeping too low a level of capital “because times are good” has led to numerous bankruptcies.
- Access to liquidity is key to survival. Having enough capital increases odds of continued access to liquidity.

**You got to know when to hold ‘em, know when to fold ‘em. Know
when to walk away and know when to run.”**

Kenny Rogers

Module VI: Organizational Risk Management

**Contemporary Financial Intermediation, Chapter 4.
Bank Risks**

KEY QUESTIONS

- What kinds of risks do organizations face?
- How are these risks interrelated?
- How should we manage these risks?
- What is Enterprise Risk Management (ERM)?
- What are the benefits of ERM?

ENTERPRISE RISK MANAGEMENT (ERM)

- The different types of risks discussed earlier are all typically correlated. This makes it inefficient to manage these risk individually. Rather, an integrated approach is called for.
- ERM is an integrated approach. It is a comprehensive and systematic approach to the management of material risks facing the organization.
- Critical to ERM is creating a categorization/taxonomy of risks for the organization

Enterprise Risk Management (ERM)

- The OCC's risk taxonomy is as follows:

The OCC's Risk Taxonomy	
Credit risk	Risk of borrower's failure to comply with loan contract terms (more detailed discussion appears in Chapter 7).
Interest rate risk	Exposure to losses arising from duration imbalances on assets and liabilities (more detailed discussion appears in Chapter 5).
Liquidity risk	Inability to realize full value on an asset owing to the need to monetize it on short notice (more detailed discussion appears in Chapter 6).
Price risk	Risk of assets varying in market value
Operational risk	Risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events.
Compliance risk	Failure to adequately monitor and supervise, to ensure compliance with laws and regulations, both public and private, applicable to the organization
Strategic risk	Failure to achieve the goals of the enterprise's articulated strategy
Reputation risk	Risk of loss of brand equity or organizational credibility that jeopardizes the organization's ability to function successfully.

Which Risks should the Institution take and which should it Lay Off or Hedge?

- Core Risks – those whose management is aligned with the core competence of the institution – are risks that should be taken. Other risks should be hedged.

Institution	Core Risk
Commercial Bank	Credit or counterparty credit risk
Investment Bank	Market or price risk
Life insurer	Mortality/ Morbidity/ Longevity risk
Property/ Casualty insurer	Assorted physical or moral hazard

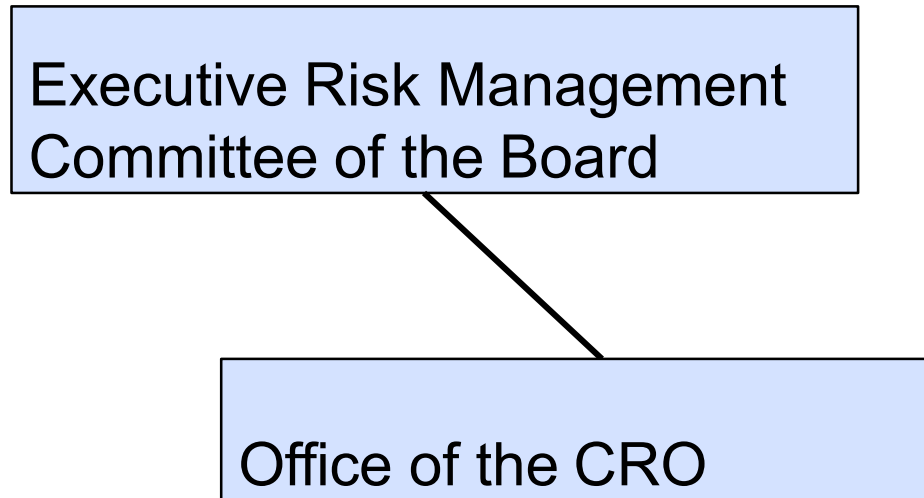
ERM and Tail Risks

- Tail risks (black swans) can be existential threats to the institution.
- Often, when tail risk events materialize, random variables that typically exhibit low correlation end up being highly correlated, and diversification fails.
- Because tail risks involve events that occur with very low probabilities, they tend to be ignored or mispriced.

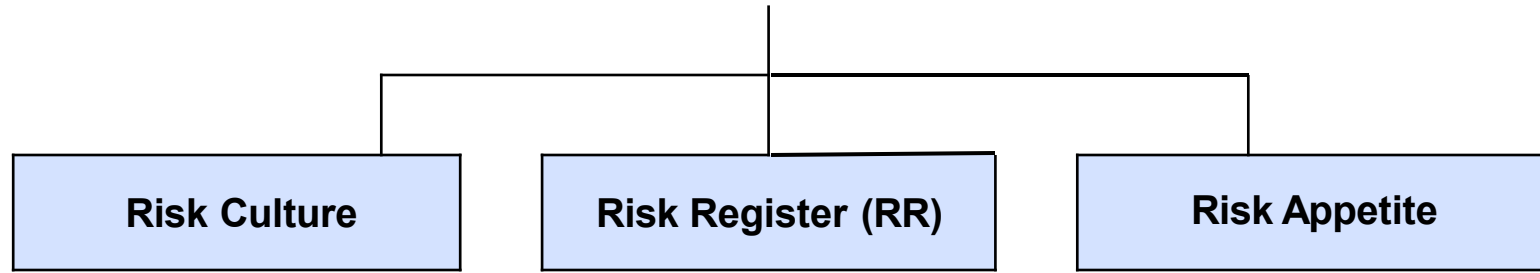
Reasons Why Tail Risks are often Ignored

- Reasons are behavioral biases:
 - ➔ Overconfidence (that a low-probability event will never occur because it has not occurred in decades)
 - ➔ Anchoring (on certain salient good events in recent experience)
 - ➔ Group think (“no sensible person” believes otherwise)
 - ➔ Discernibility (difficulty in distinguishing indefinitely small probabilities from zero).
 - ➔ Confirmation bias (tending to interpret, favor, and recall information in a way that confirms one’s pre-existing beliefs and disregard contradictory signals, e.g., recent elections votes in the U.K. and the U.S.)

ORGANIZING FOR ERM



ERM Communication Tools



- Shared values and practices
- Implicit contracts and (often) unwritten rules of conduct

- A record of material risks facing the organization that need to be monitored.
- Recording of exposures in RR with describe each risk, and include estimates of magnitudes and probabilities of occurrence.

- Written statement of risk preferences and tolerances for risk.
- Complements RR.

Conclusion

- For any organization, the major types of risks are market risk, liquidity risk, reputation risk, and operational and other risks.
- These risks are interrelated, depending in an important way on the presence of asymmetric information.
- Enterprise Risk Management (ERM) is a way to holistically manage these risks.
- These risks can be managed (as part of ERM) but not totally avoided.

Group Exercise #3: Hydro Case

- What is Hydro One's strategy and what risks does the company face?
- Putting yourself in the shoes of CEO Laura Formosa and the Hydro One Board, what risks does Hydro One face?
- What difference does ERM make to the running of a business, compared to not having a process like this at all?
- As you consider Hydro One's three-phase ERM process, what do you see as its strengths and weaknesses?
- What are your key takeaways?

RECENT GLOBAL BANKING AND MARKET DEVELOPMENTS

- Dramatic reduction in brick-and-mortar branches
- Emergence of depositories called “neo-banks” that have no physical branches
- Checkless banking system (e.g., Netherlands)
- Growth of digital banking as a replacement for in-person banking (e.g., Sweden)
- Blurring of boundaries between banks and markets

RECENT GLOBAL BANKING And MARKET DEVELOPMENTS (CONT'D.)

- Non-banks beginning to take market share away from banks in some markets (e.g., mortgage lending in the US)
- Decline of relationship lending (based on hard as well as soft information) and dominance of transaction lending with hard information
- Increasing reliance of central banks on expanding deposit insurance coverage (e.g., TAG in US and unlimited coverage in Denmark during after GFC) and bailing out failing banks

Module VI:

Banking and Higher Purpose

HIGHER PURPOSE

- Should a bank have a higher purpose?
- If yes, what should it be?
- Would the pursuit of HP increase the bank's risk?
- Would bank regulators tolerate it?

BANK OF BIRD-IN-HAND

CASE STUDY: GROUP EXERCISE #4

- After watching the video of Bank of Bird in hand in class, please answer Questions 1, 2 and 3 in the case study.
- Put here link to video

Module VII:

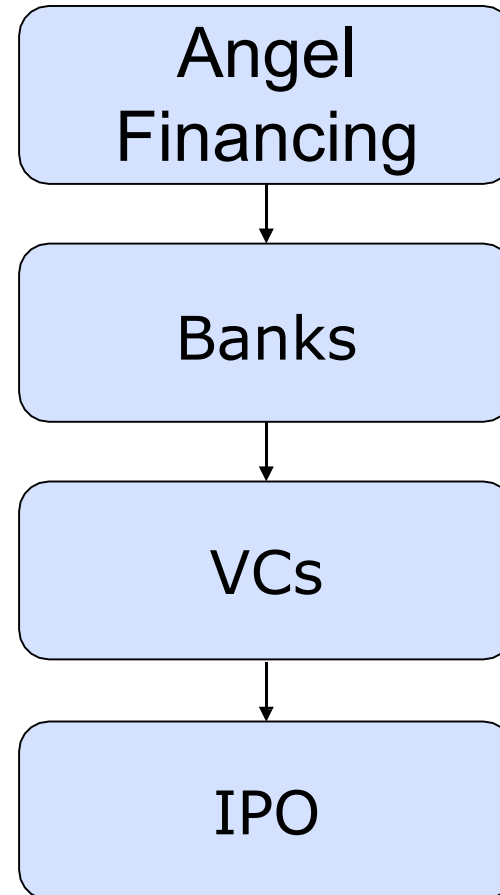
Raising Financing and Emerging Market Opportunities

HIERARCHY OF FINANCING SOURCES

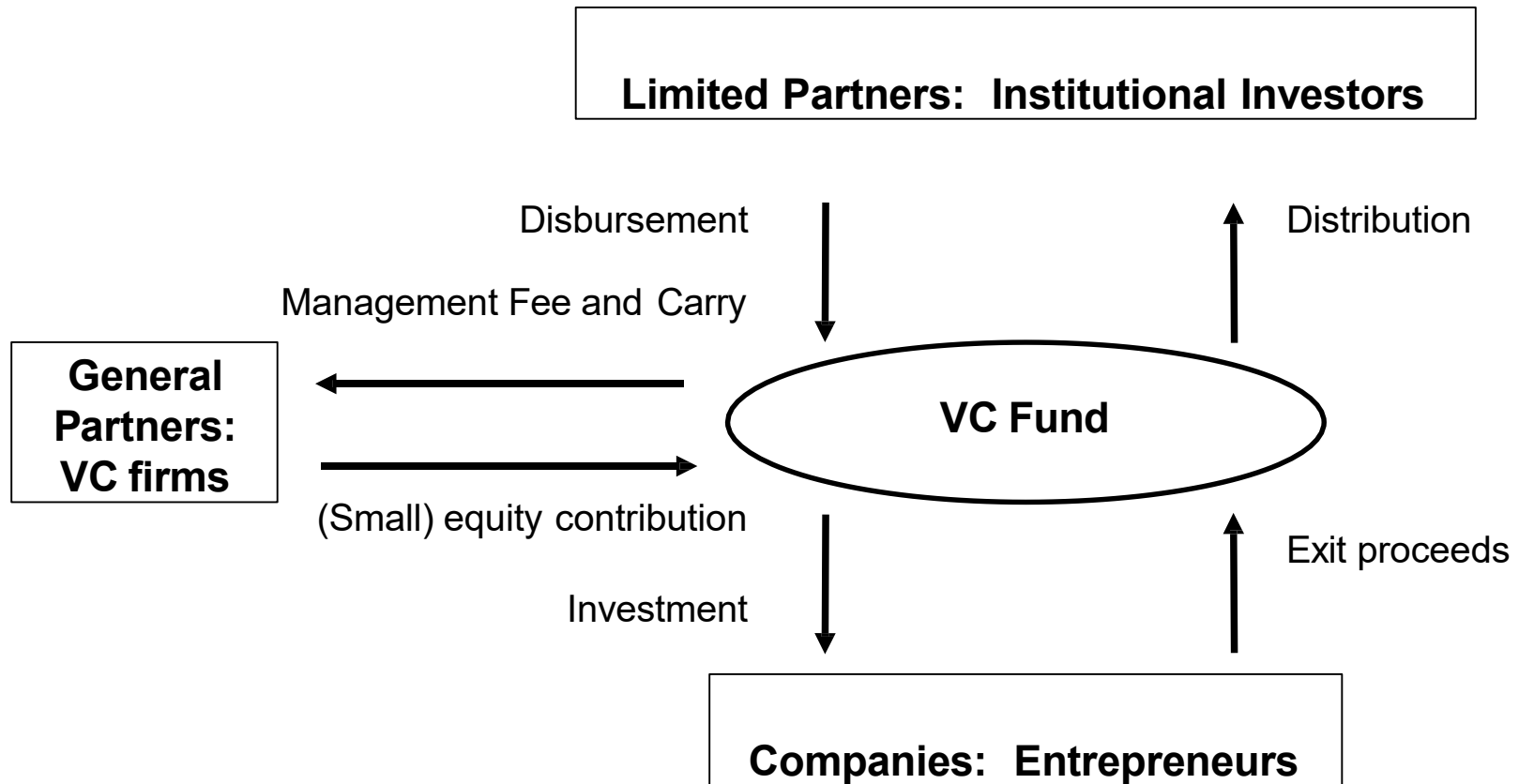
- IPOs are underpriced.

➔ Why?

- ❖ Litigation risk
- ❖ Branding
- ❖ Adverse Selection



How Does Venture Capital Work?



SOME FACTS

- VC-backed companies account for 5.3%–7.3% of employment in the U.S.
- VC investments in companies account for 0.15% of the total value of the stock market
- Returns to VC funds are comparable to those of overall stock market
- Venture capital accounts for about 15% of industrial innovations.
- Only 1/6% of new businesses receive VC funding

- But...60% of companies going public are VC-funded: Microsoft, apple, google, cisco, eBay, amazon, Starbucks, adobe, intuit, etc.
- Other capital sources: Private equity; preferred stock issues; subordinated debt
 - ➡ Economic tradeoffs: liquidity/capital access vs. ownership and control rights
 - ➡ Another issue: operating flexibility for strategic decisions (will external financiers always agree with your preferred investments?)

VC SERVICES / ACTIVITIES

Screening and Selecting Deals

Factors

- Market size
- Strategy
- Technology
- Customer adoption
- Competition
- Management team

Sophisticated Contracting and Structuring of Their Investments

- VC avoids taking control if entrepreneur does well but has potential to gain control if things go poorly

Improve Outcomes of and Add Value to Portfolio Companies by Providing Governance and Monitoring

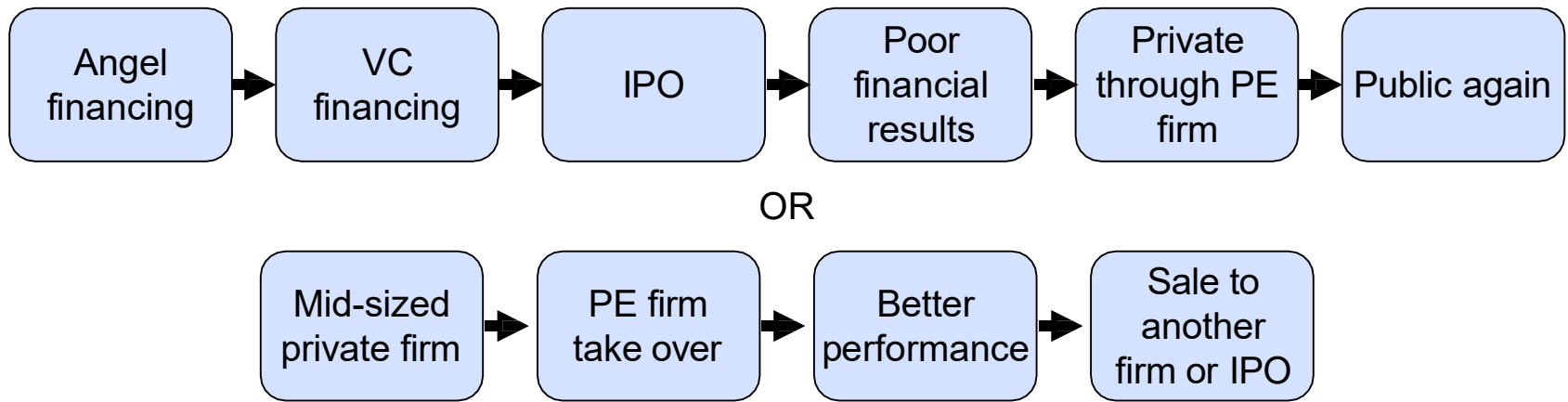
Key Factors In How VC's Value New Ventures

(1 least important, 5 most important)

		Score
1.	Quality of Management	4.5
2.	Size of market	3.8
3.	Product qualities	3.7
4.	Rate of market growth	3.5
5.	Competition	3.5
6.	Barriers to entry	3.4
7.	Company stage of development	3.2
8.	Industry company is in	3.0

Private Equity

- While VCs provide equity funding to start-ups and also companies further along on the development scale, eventually taking them public through IPOs, *PE firms* (i) invest equity in developed small and mid-sized firms, often take complete ownership, professionalize management and improve operations; and (ii) also do the same in large public firms, taking them private.



WHERE DOES FINTECH FIT IN?

- See Chapters 5 and 11 in The Purpose of Banking.
- Fintech is part of increased banking competition and use of technology to improve productivity. It involves bypassing trusted intermediaries. It lowers the cost of intermediation. But it requires people to trust opaque technology and motives of code writers. A big scandal can set it back.

WHAT STUCK?

- Pair up with the person next to you and summarize your top 3 takeaways. List any remaining question you would like to ask.

END OF COURSE